

Encoder-Decoder models and text-to-text generation

NLP Week 10

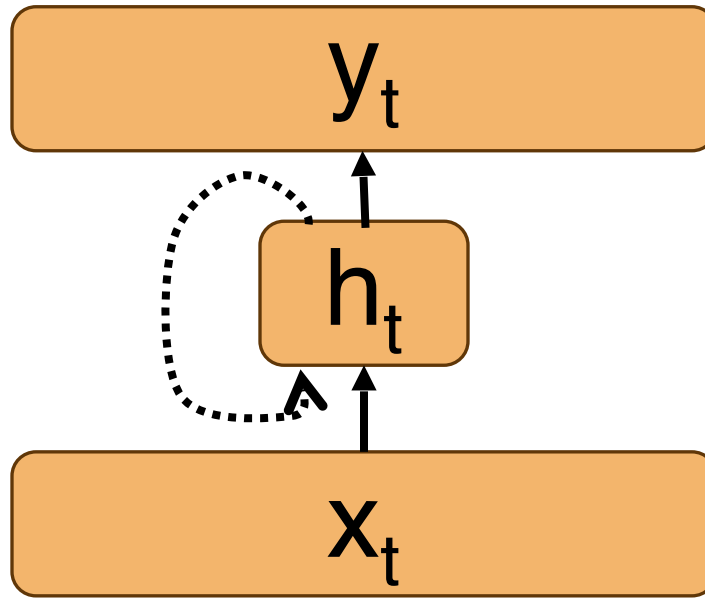
Thanks to Dan Jurafsky and Marius Mosbach for some of today's slides

Plan for today

1. **Introducing Sequence to sequence (seq-to-seq) with encoder-decoder RNNs**
2. **Encoder-decoder Transformers**
3. **Machine Translation**
4. **T5: text-to-text everywhere**
5. ***Group exercise : case study***

Simple recurrent neural units (SRN or Elman Net)

Recall from Week 6



The hidden layer has a recurrence as part of its input

The activation value h_t depends on x_t but also h_{t-1} !

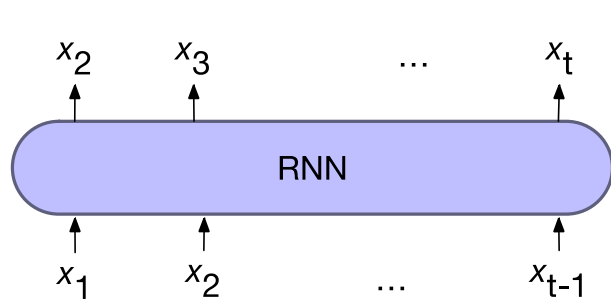
RNN Architectures

We will see three types of RNN-based architectures that can be used for different tasks:

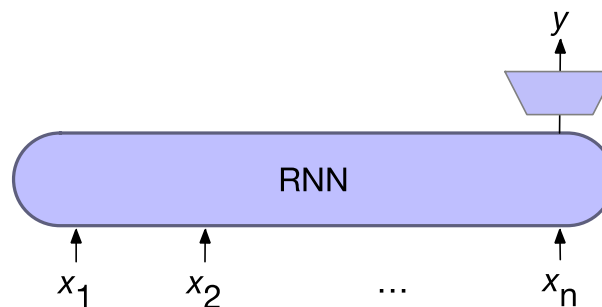
1. RNNs for language modeling

2. RNNs for classification

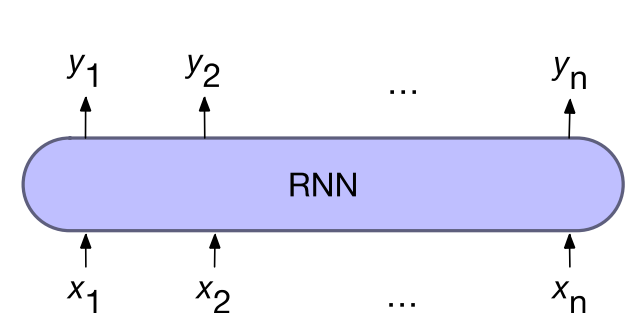
3. RNNs for sequences



c) language modeling



b) sequence classification

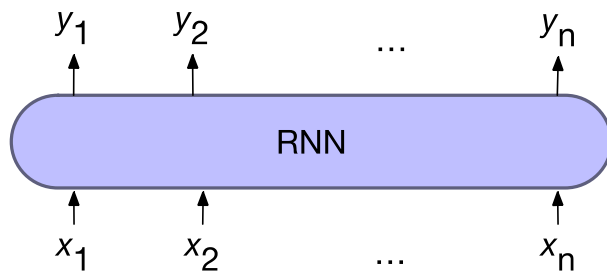


a) sequence labeling

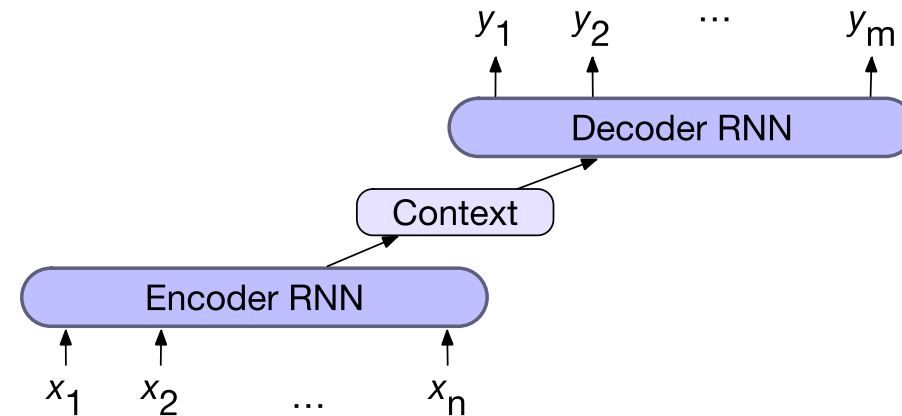
Two types of sequence prediction tasks

Recall from Week 6

1. **sequence labeling** : When output sequence length matches input
2. **seq-to-seq or encoder-decoder** : When output and input sequence are of different lengths



a) sequence labeling

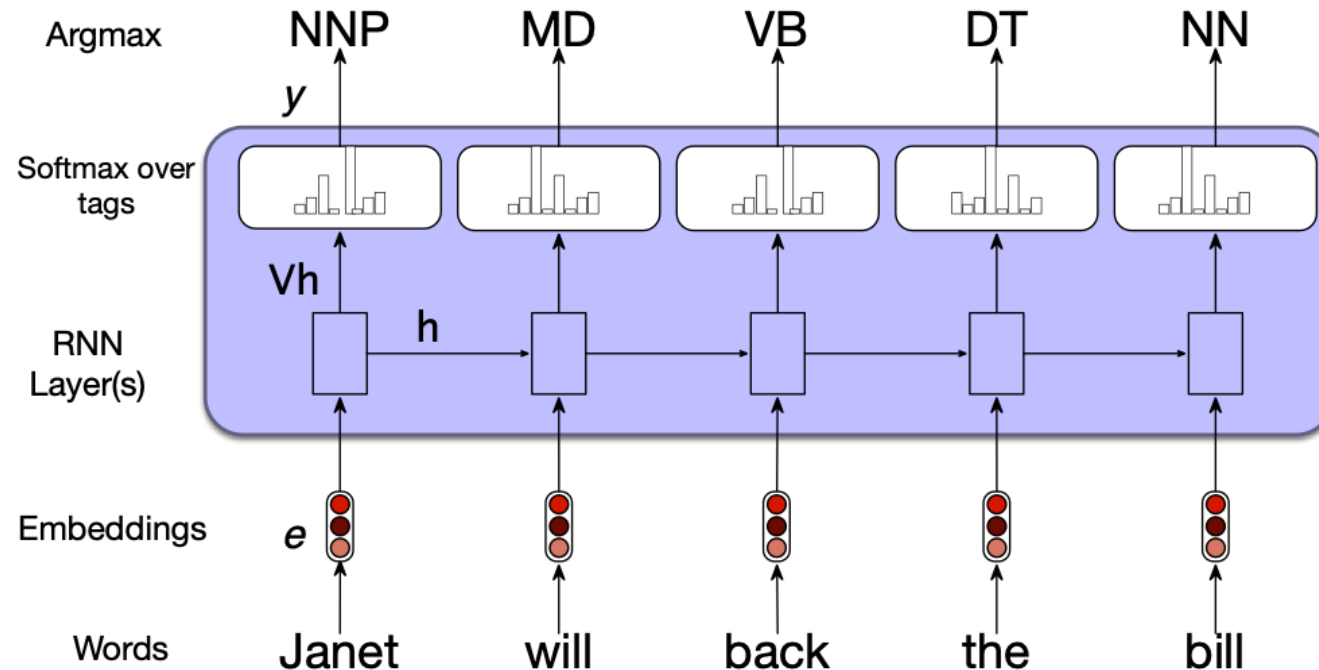


d) encoder-decoder

RNNs for sequence labeling

Assign a label to each element of a sequence (eg. POS tagging):

- The main difference from LM architecture and classification architecture is that the output dimension for output layer is over labels (not vocab), and there is an output prediction at each time step (not just the last one).

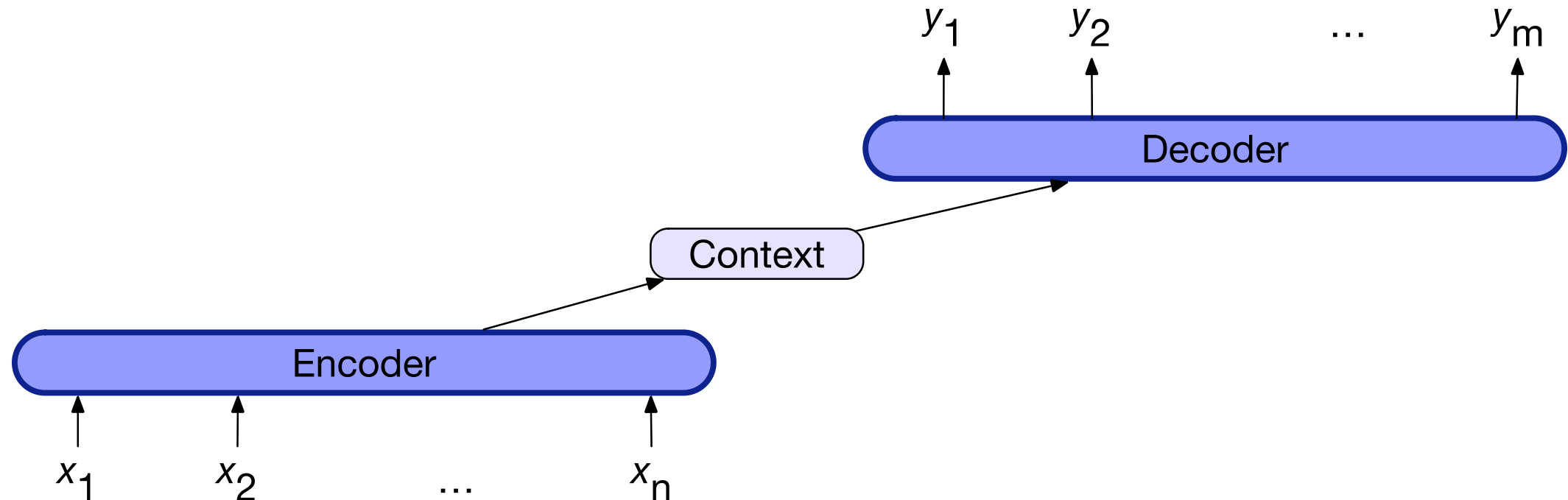


RNNs for sequence to sequence tasks

An example of such a task is translation:

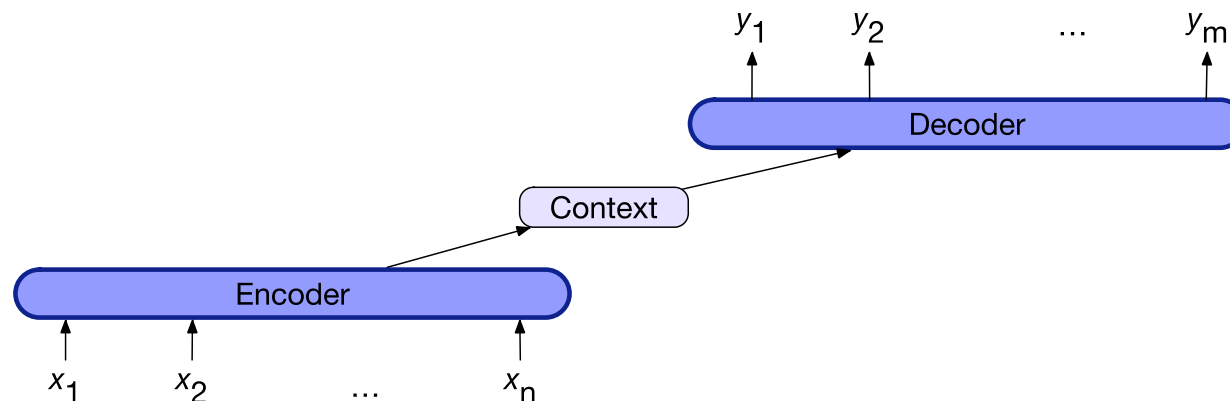
‘Have you tasted *loroco* before?’

‘Avez-vous déjà goûté au *loroco*?’



3 components of an encoder-decoder

1. An encoder that accepts an input sequence, $x_{1:n}$, and generates a corresponding sequence of contextualized representations, $h_{1:n}$.
2. A context vector, c , which is a function of $h_{1:n}$, and conveys the essence of the input to the decoder.
3. A decoder, which accepts c as input and generates an arbitrary length sequence of hidden states $h_{1:m}$, from which a corresponding sequence of output states $y_{1:m}$, can be obtained



Encoder-decoder for translation

Regular language modeling

$$p(y) = p(y_1)p(y_2|y_1)p(y_3|y_1, y_2) \dots p(y_m|y_1, \dots, y_{m-1})$$

Conditioned sequence scoring

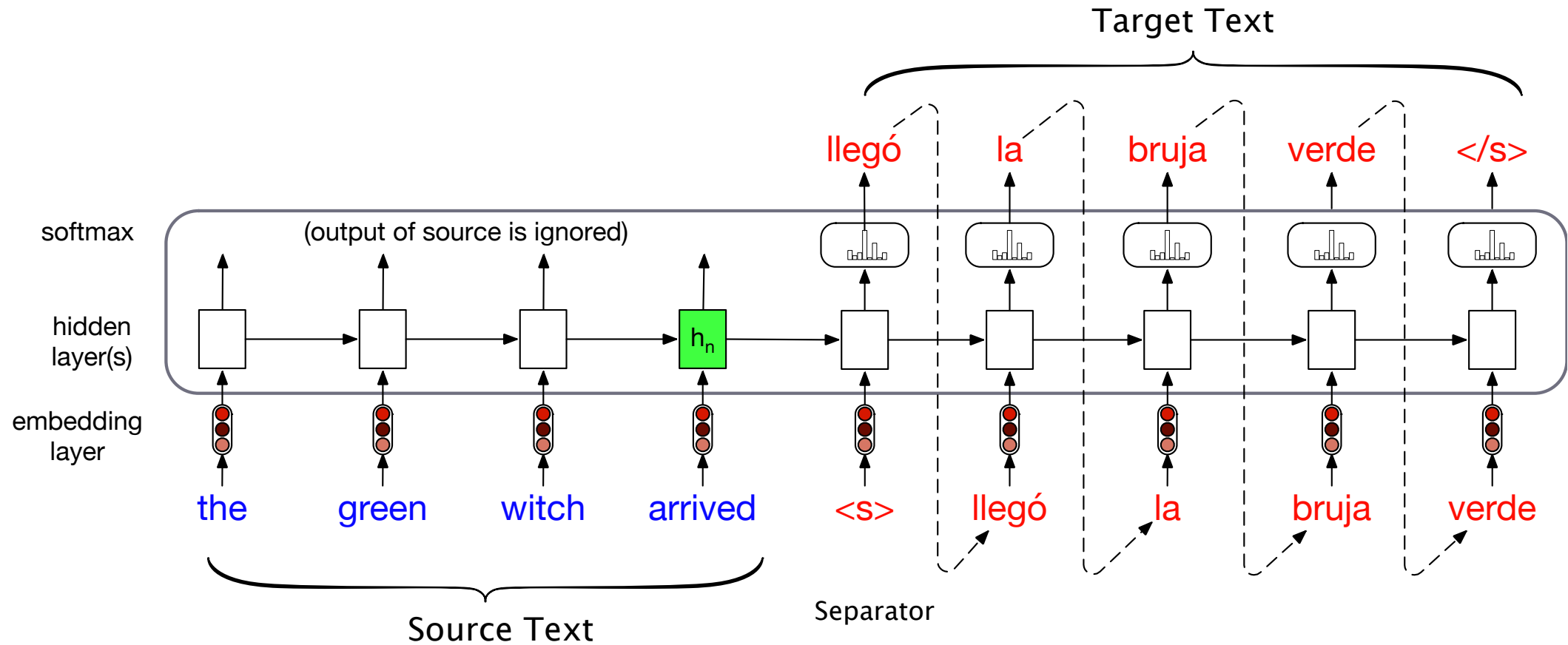
Let x be the source text plus a separate token $\langle s \rangle$ and y the target

$x = \text{The green witch arrive } \langle s \rangle$

$y = \textit{llego' la bruja verde}$

$$p(y|x) = p(y_1|x)p(y_2|y_1, x)p(y_3|y_1, y_2, x) \dots p(y_m|y_1, \dots, y_{m-1}, x)$$

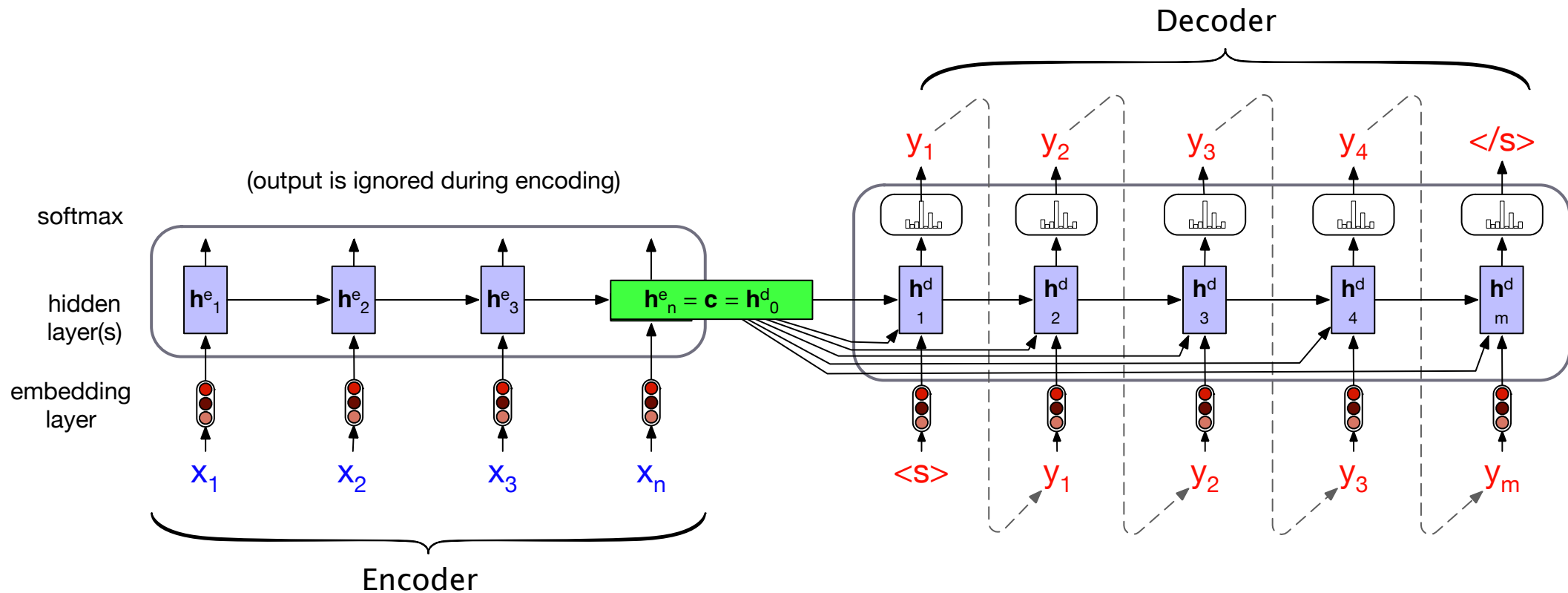
Encoder-decoder simplified



Encoder-decoder showing context

Where each hidden state of the decoder is conditioned not only on the input and prior hidden state, but also the context.

$$\mathbf{h}_t^d = g(\hat{y}_{t-1}, \mathbf{h}_{t-1}^d, \mathbf{c})$$



Encoder-decoder equations

$$\begin{aligned}\mathbf{c} &= \mathbf{h}_n^e \\ \mathbf{h}_0^d &= \mathbf{c} \\ \mathbf{h}_t^d &= g(\hat{\mathbf{y}}_{t-1}, \mathbf{h}_{t-1}^d, \mathbf{c}) \\ \hat{\mathbf{y}}_t &= \text{softmax}(\mathbf{h}_t^d)\end{aligned}$$

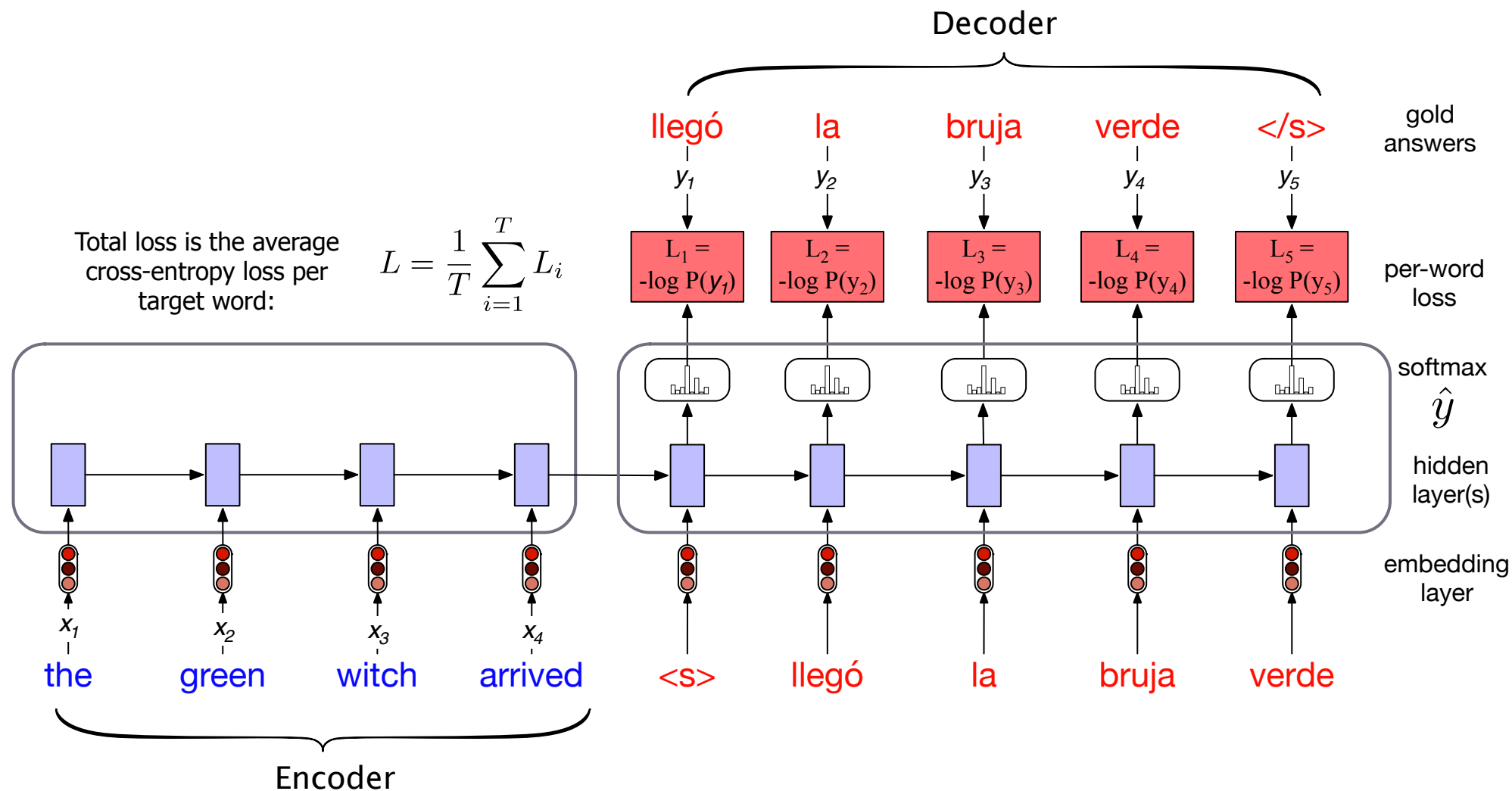
g is a stand-in for some flavor of RNN

$\hat{\mathbf{y}}_{t-1}$ is the embedding for the output sampled from the softmax at the previous step

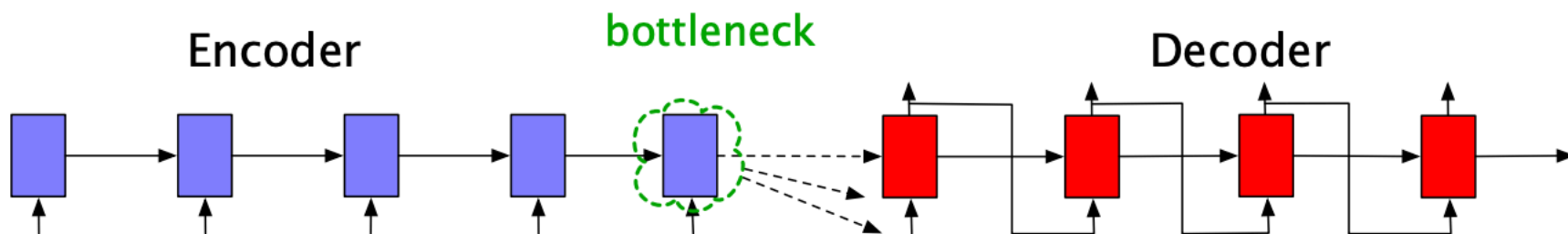
$\hat{\mathbf{y}}_t$ is output vector of probabilities over vocab, representing the probability of each word occurring at time t .

To generate text, we sample from this distribution $\hat{\mathbf{y}}_t$.

Training the encoder-decoder with teacher forcing

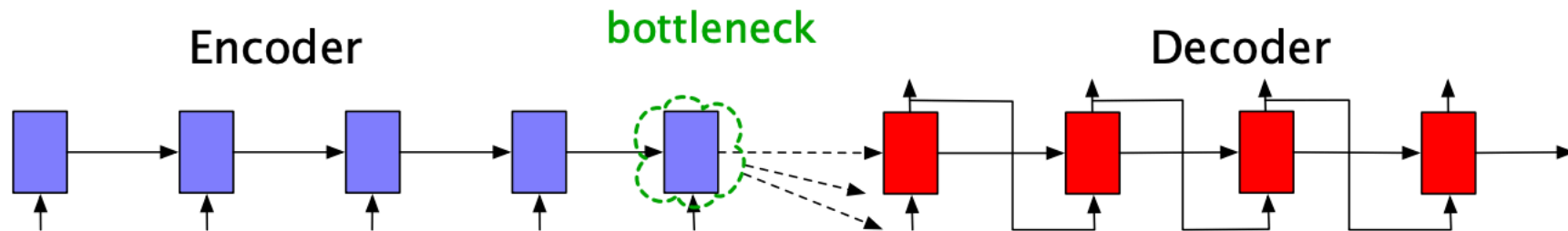


RNN enc-dec and simplified attention



RNN enc-dec and simplified attention

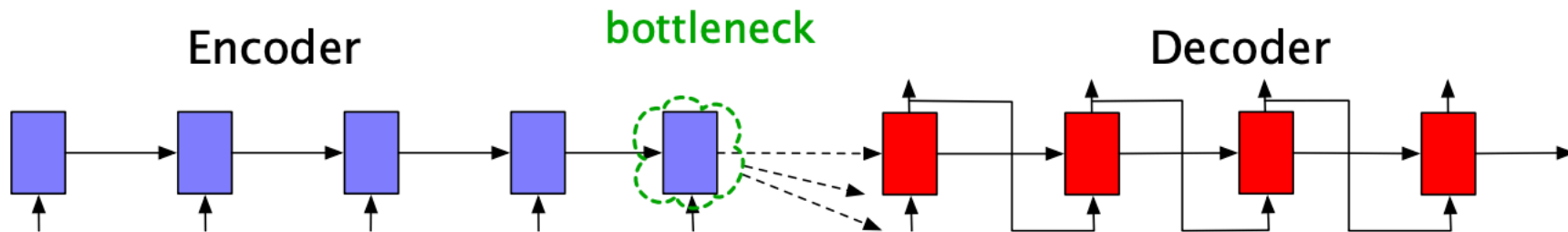
You may have noticed a problem...



RNN enc-dec and simplified attention

You may have noticed a problem...

c the context vector is taken from the final state of the encoder, creating a bottleneck, where **c** must encode useful information about the reference sentence for all of the words in the target sentence.

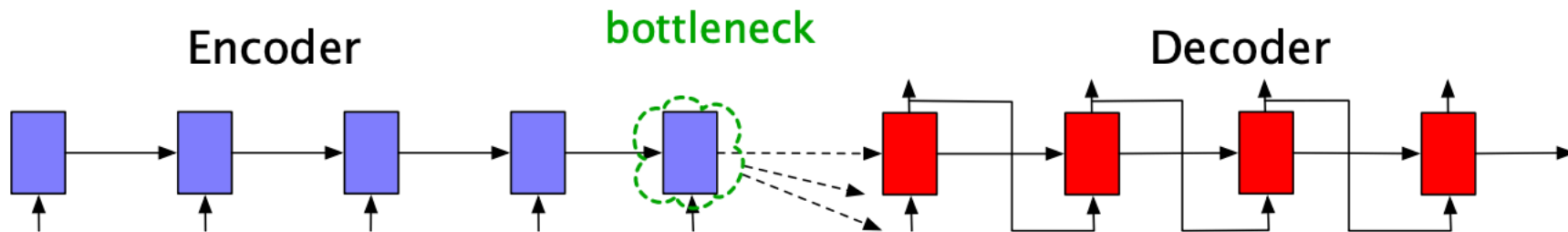


RNN enc-dec and simplified attention

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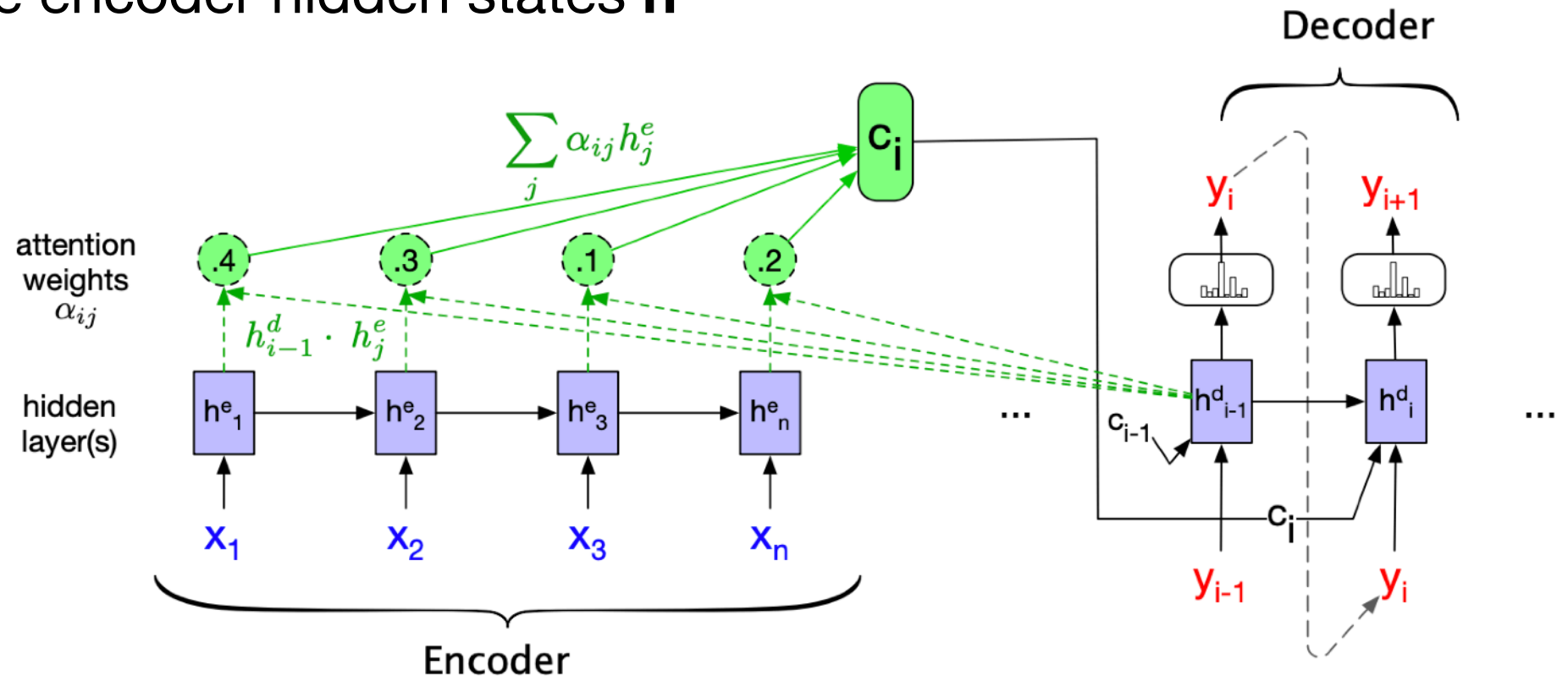
Solution: *attention!* (...where it all started)



RNN enc-dec and simplified attention

Each step i of the decoder now gets as input the $\mathbf{y}_{i-1}$ embedding for the previous word and vector $\mathbf{c}_i$, which “attends to” information from all of the encoder hidden states $\mathbf{h}^e$

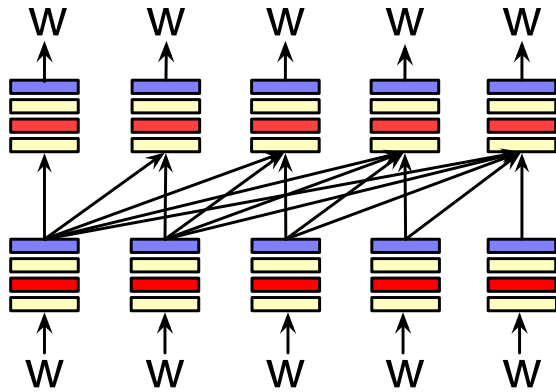
$$\mathbf{c}_i = \sum_j \alpha_{ij} \mathbf{h}_j^e$$



Encoder-decoder Transformers (... “attention is all you need”)

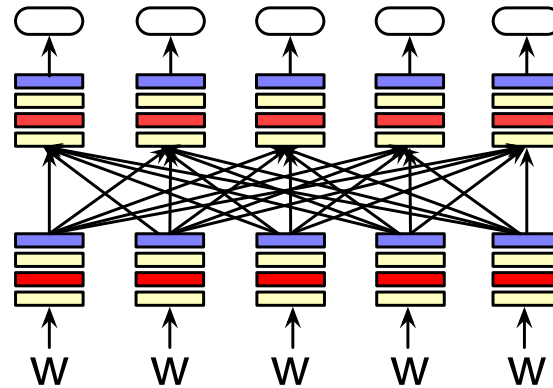
- Now we will see a different class of Transformer-based models that have been / are widely used

Three architectures for large language models



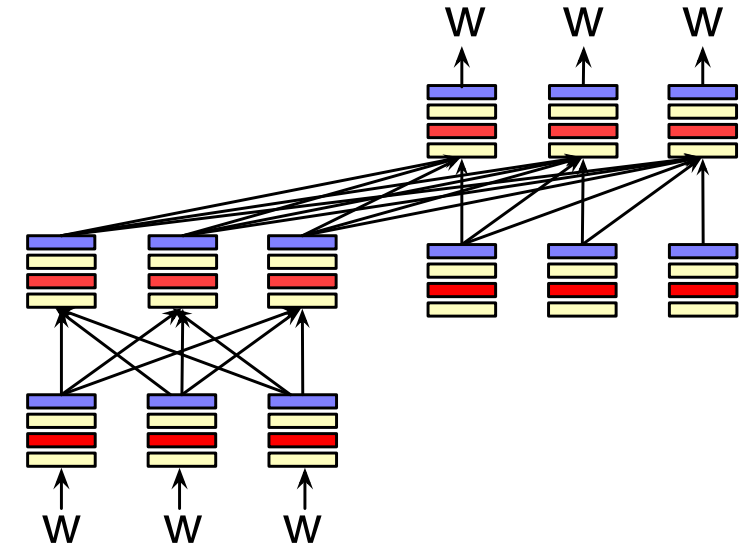
Decoders

GPT, Claude,
Llama
Mixtral



Encoders

BERT family,
HuBERT

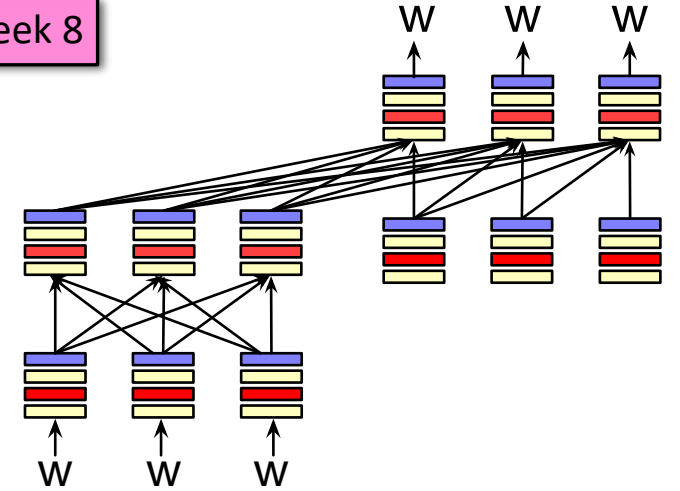


Encoder-decoders

Flan-T5, Whisper

Encoder-Decoders

Recall from Week 8



- Trained to map from one sequence to another
- Very popular for:
 - machine translation (map from one language to another)
 - speech recognition (map from acoustics to words)

The paper that started it all

Recall from Week 7

Transformer: a specific kind of network architecture, like a fancier feedforward network, but based on attention

Attention Is All You Need

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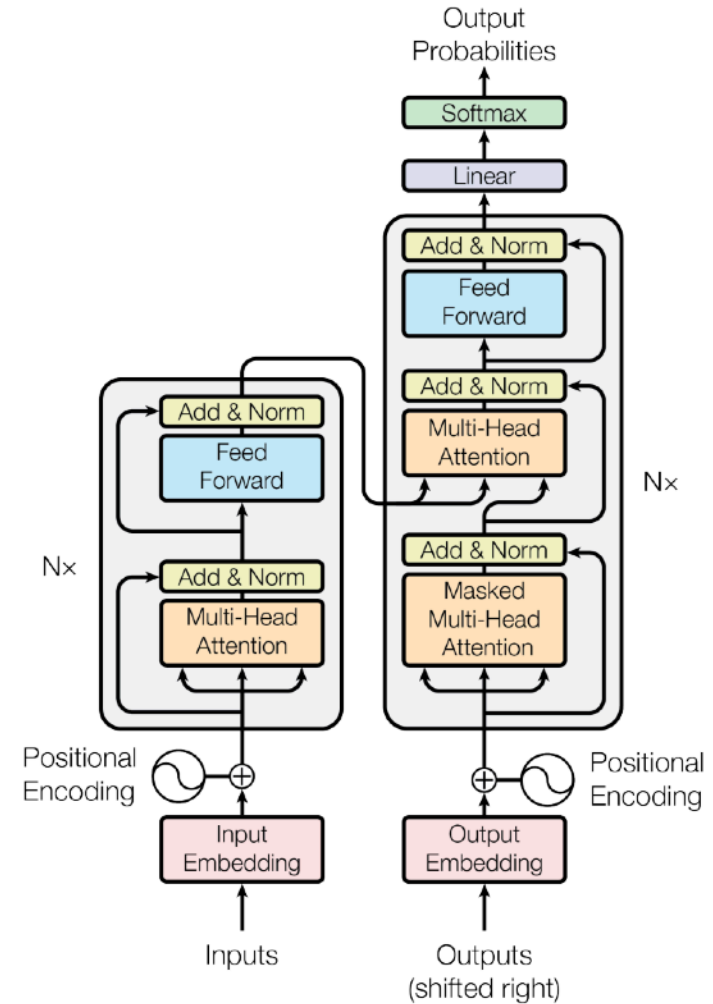
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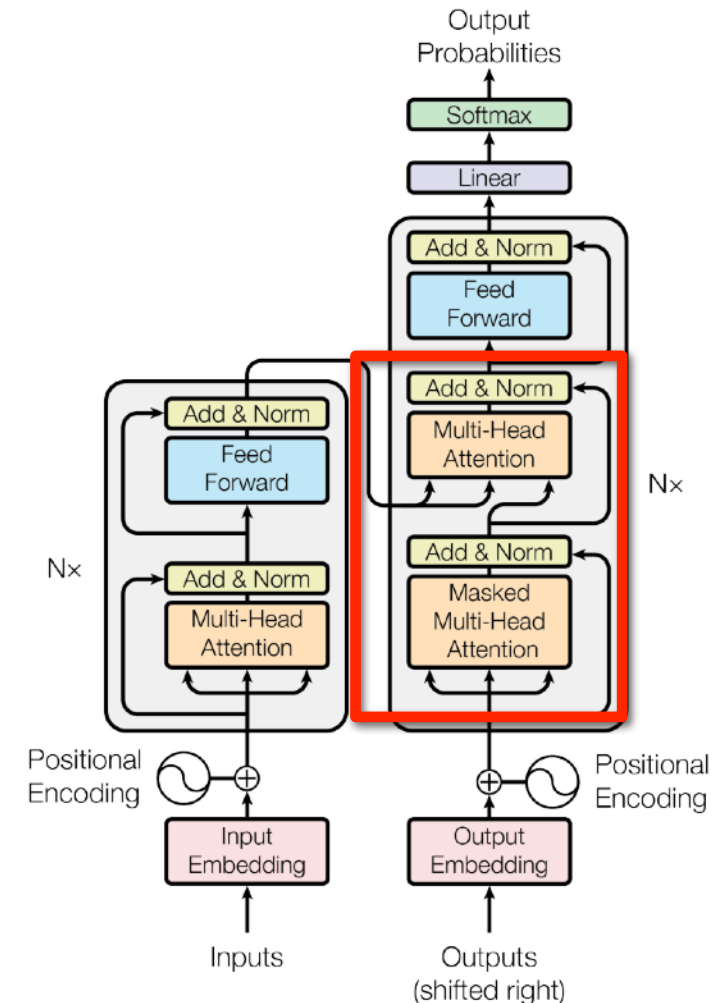
The original transformers was an enc-dec model

- **Encoder**
 - Bidirectional self-attention
- **Decoder**
 - Causal self-attention
- How do encoder and decoder “communicate” with each other?
 - **Cross-attention**

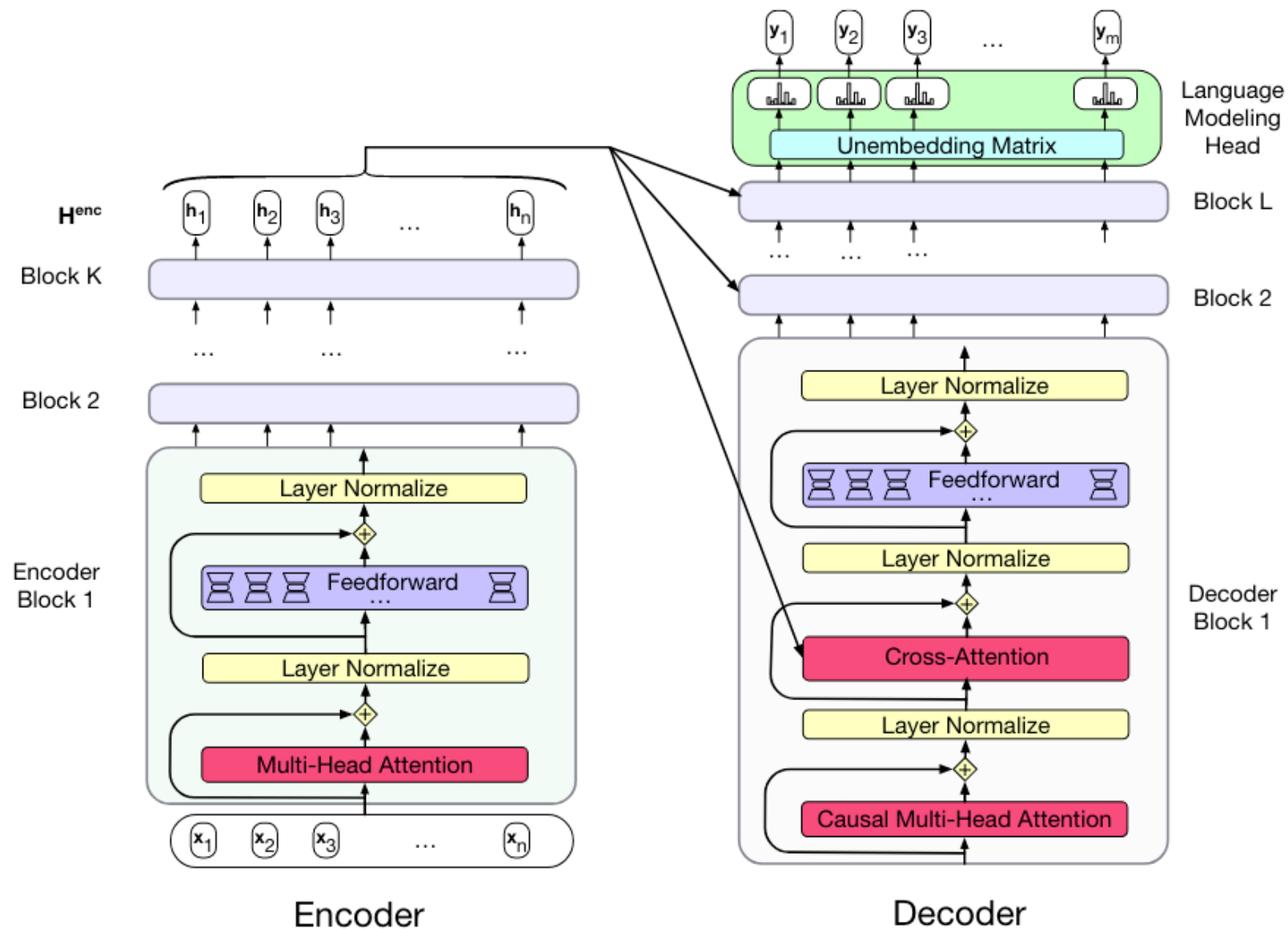


Cross-attention in the decoder

- The transformer block of the decoder has two attention sub-layers instead of just one!
- **Self-attention** with $\mathbf{Q} = \mathbf{XW}^Q$, $\mathbf{K} = \mathbf{XW}^K$, and $\mathbf{V} = \mathbf{XW}^V$
- **Cross-attention** with $\mathbf{Q} = \mathbf{XW}^Q$, $\mathbf{K} = \mathbf{X}_{\text{enc}}^L \mathbf{W}^K$, and $\mathbf{V} = \mathbf{X}_{\text{enc}}^L \mathbf{W}^V$
- Where keys and values are based on the representations from the last encoder layer!



The model architecture



Machine Translation via enc-dec models

TextBilderDokumenteWebsites

Sprache erkennenFriesischDeutschEnglisch

HEC Montréal (French: Hautes études commerciales de Montréal; English: High Commercial Studies of Montreal) is a bilingual public business school located in Montreal, Quebec, Canada. Founded in 1907, HEC Montréal is the graduate business school of the Université de Montréal and is the first established school of management in Canada.[2][3]

HEC Montréal offers undergraduate, graduate, and post-graduate programs, including Bachelor of Business Administration (BBA), Master of Science in Administration (MSc), Master of Management (MM), Master of Business Administration (MBA), and PhD in Administration, in addition to a joint Executive MBA program with McGill University.



675 / 5.000

↔Französisch (Kanada)FriesischEnglisch

HEC Montréal (en français : Hautes études commerciales de Montréal ; en anglais : High Commercial Studies of Montréal) est une école de commerce publique bilingue située à Montréal, Québec, Canada. Fondée en 1907, HEC Montréal est l'école supérieure de gestion de l'Université de Montréal et la première école de gestion établie au Canada.[2][3]

HEC Montréal offre des programmes de premier, deuxième et troisième cycles, dont un baccalauréat en administration des affaires (B.A.A.), une maîtrise en sciences de l'administration (M.Sc.), une maîtrise en gestion (M.M.), une maîtrise en administration des affaires (M.B.A.) et un doctorat en administration, en plus d'un programme conjoint de MBA pour cadres avec l'Université McGill.





Feedback geben

Language diversity

- There are ~ 7000 languages in the world
- Some aspects of human language seem to be *universal*
 - Holding true for every language
 - *Statistically universal*: true for almost all languages
- E.g, every language seems to have nouns and verbs
- Yet, languages also **differ in many ways**
- This makes MT an interesting and challenging problem for language technology and research

Typology

- Sometimes differences between languages are systematic
- **Typology** studies these systematic cross-linguistic similarities and differences
- Example:
 - **Word order**
 - SVO
 - SOV
 - VSO
 - Who can think of example languages?



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- Example:
 - **Word order**
 - SVO: French, English, German
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 - Who can think of example languages



English: *He wrote a letter to a friend*
Japanese: *tomodachi ni tegami-o kaita*
 friend to letter wrote
Arabic: *katabt risāla li šadq*
 wrote letter to friend

Lexical divergences

- When translating, we also need to map individual words from one to the other
- But the appropriate word can vary depending on the *context*
- Example:
 - wall (English)
 - Mauer or Wand (German)
 - Wand is typically something inside

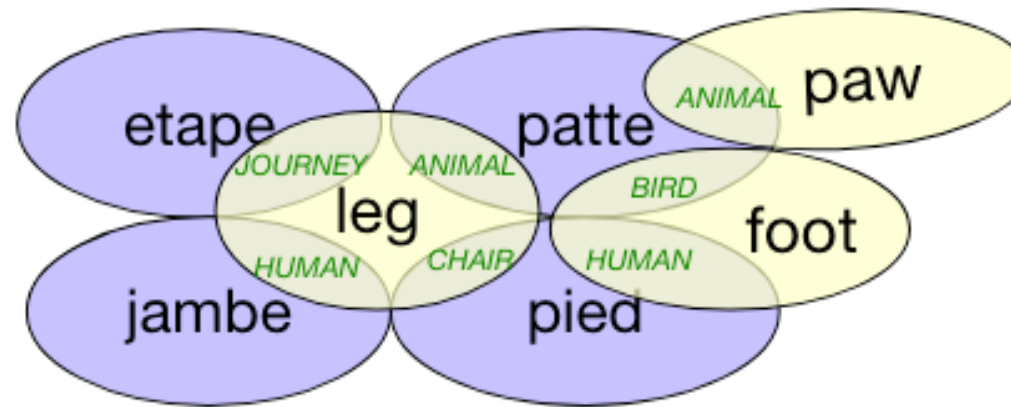


The Berlin Wall (Berliner Mauer)

Berliner Wand would sound really strange to any German

Lexical divergences

- How would you translate the English words *leg*, *paw*, or *foot* to French?



- Lexical gaps also exist : when no word or phrase can express the exact meaning in another language, eg. *Dépaysement* in French doesn't have an exact translation in English

Machine translation with enc-dec models

- We need to make some **assumptions** first
 - Often, MT systems assume that sentences are independent, i.e., we work with a corpus of **aligned** sentences in two (or more) languages
- What is your objective?
 - $P(y_1, \dots, y_m \mid x_1, \dots, x_n)$
 - Parameterized by an enc-dec neural network

$$\begin{aligned}\mathbf{h} &= \text{encoder}(x) \\ y_{t+1} &= \text{decoder}(\mathbf{h}, y_1, \dots, y_t) \quad \forall t \in [1, \dots, m]\end{aligned}$$

MT training data

E1: "Good morning," said the little prince.

F1: -Bonjour, dit le petit prince.

E2: "Good morning," said the merchant.

F2: -Bonjour, dit le marchand de pilules perfectionnées qui apaisent la soif.

E3: This was a merchant who sold pills that had been perfected to quench thirst.

F3: On en avale une par semaine et l'on n'éprouve plus le besoin de boire.

E4: You just swallow one pill a week and you won't feel the need for anything to drink.

F4: -C'est une grosse économie de temps, dit le marchand.

E5: "They save a huge amount of time," said the merchant.

F5: Les experts ont fait des calculs.

E6: "Fifty-three minutes a week."

F6: On épargne cinquante-trois minutes par semaine.

E7: "If I had fifty-three minutes to spend?" said the little prince to himself.

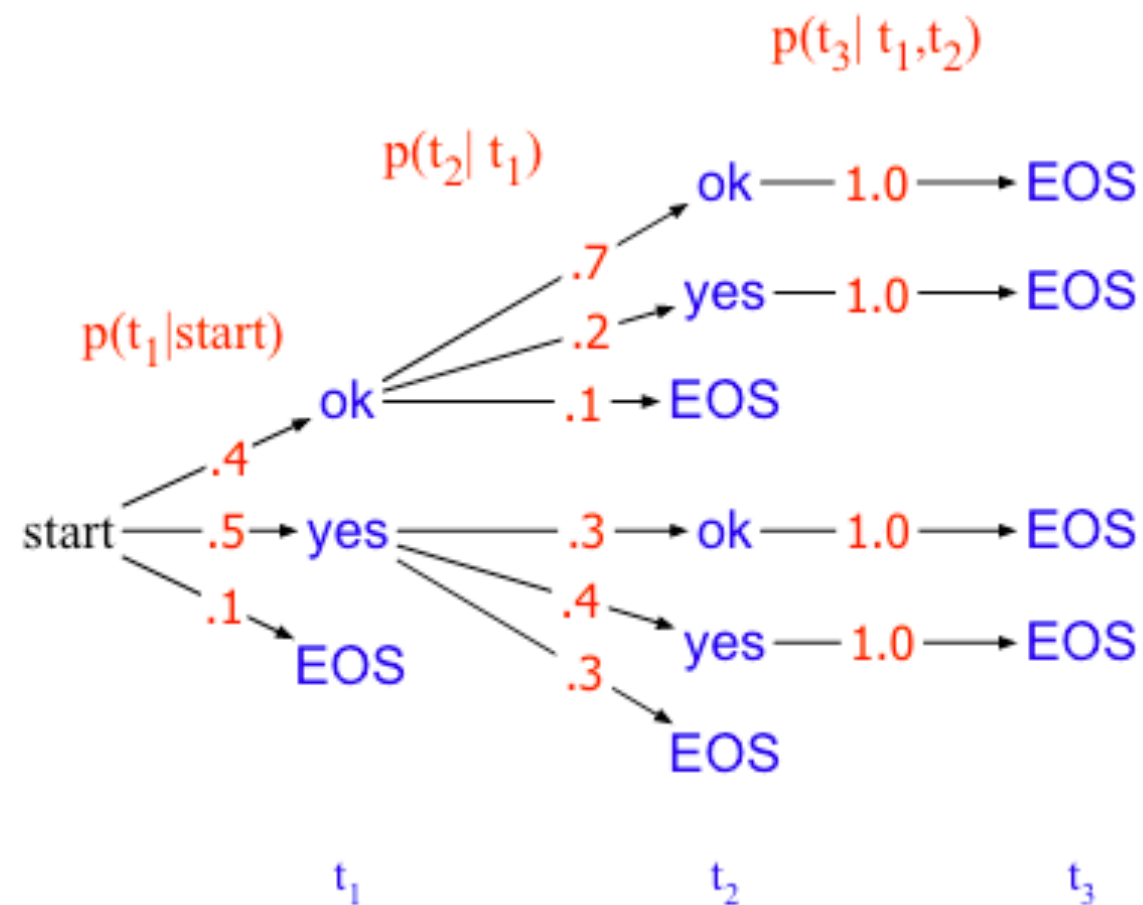
F7: "Moi, se dit le petit prince, si j'avais cinquante-trois minutes à dépenser, je marcherais tout doucement vers une fontaine..."

E8: "I would take a stroll to a spring of fresh water"

Decoding via beam-search

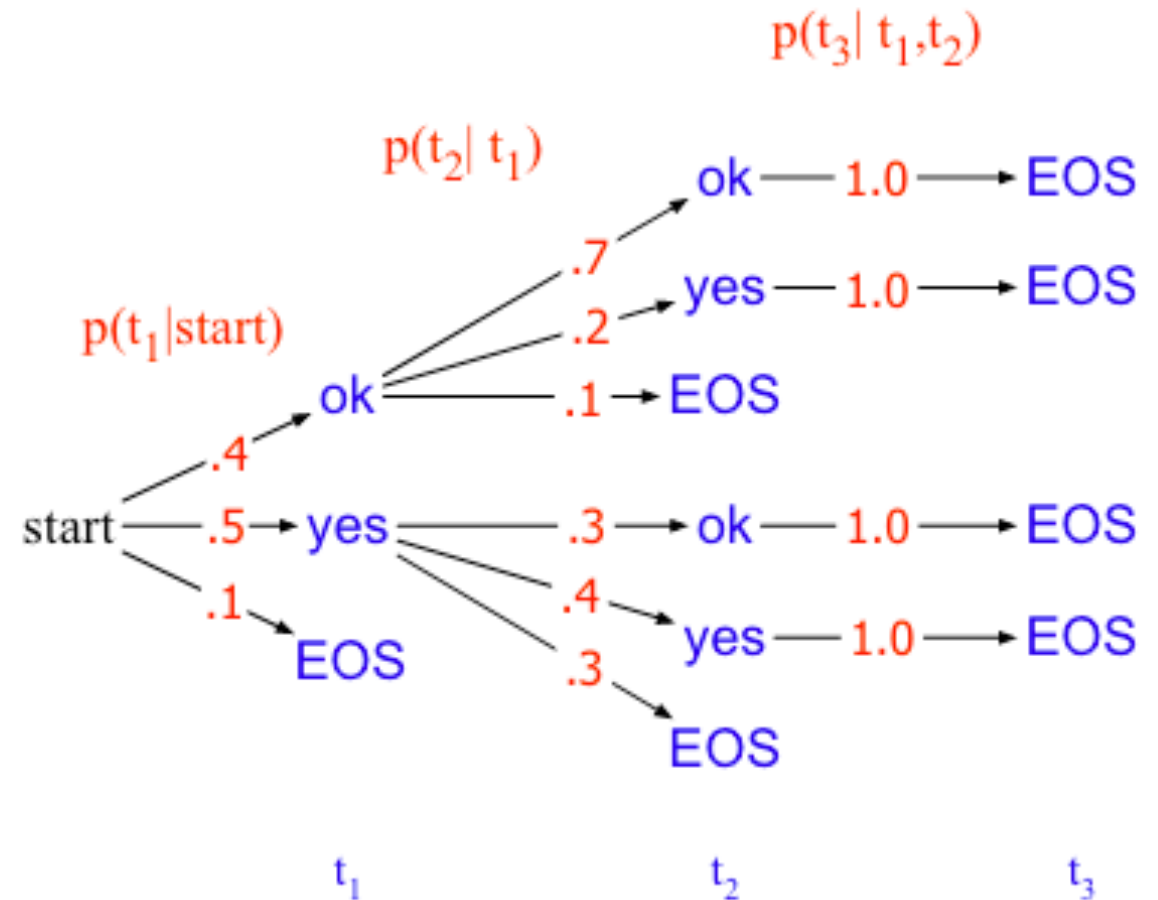
- We have seen a few decoding methods for text generation tasks, such as greedy decoding where always picked the word with the highest probability
 - $\hat{w} = \operatorname{argmax}_{w \in \mathcal{V}} P(w \mid \mathbf{w}_{<t})$
- But though greedy is +high quality it isn't necessarily optimal, especially in translation settings.
- We also saw k-top and p-top sampling which helped with +diversity for generation, but when we are translating, we don't necessarily want diversity.
- **Beam search** is an alternative method for decoding
- It considers multiple alternatives (hypotheses) in parallel to find the optimal one - it is the best choice for tasks like machine translation

Decoding via beam-search



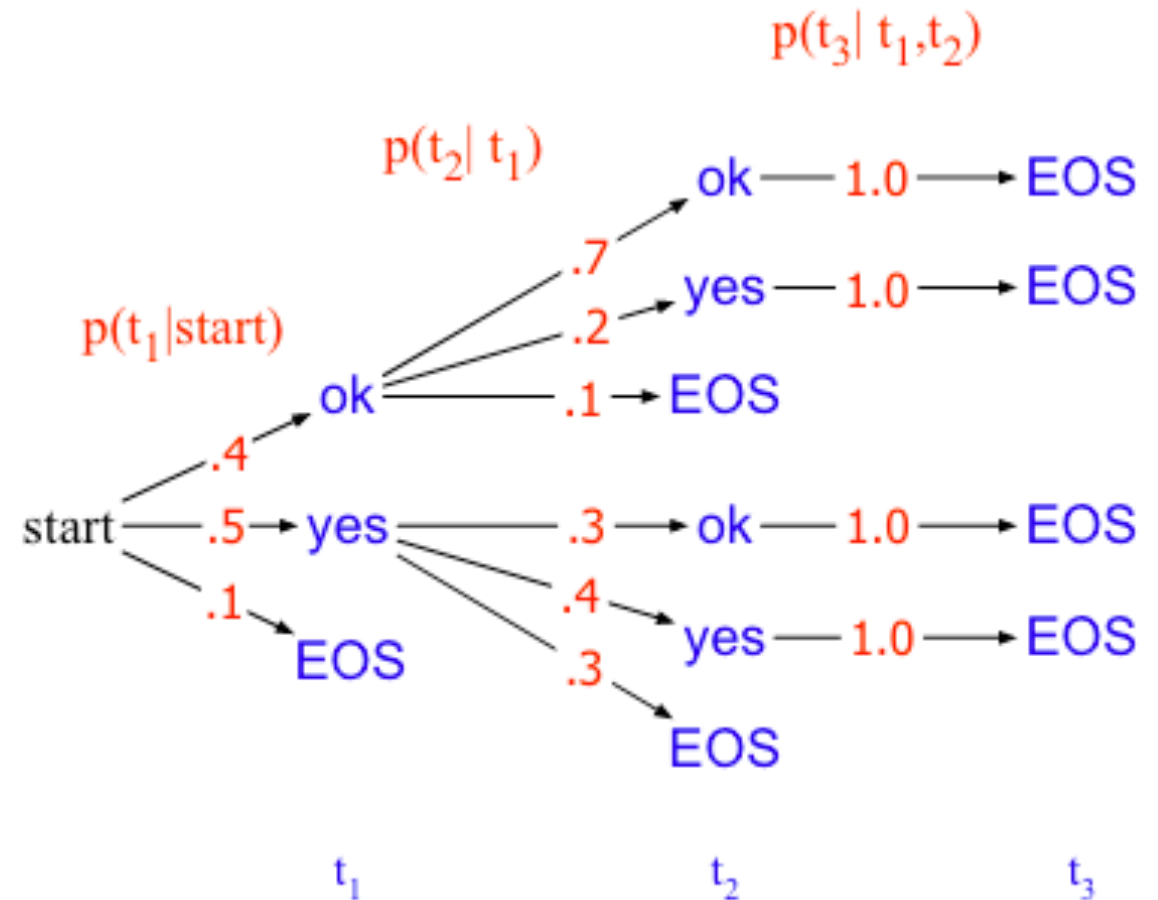
Decoding via beam-search

- In the toy problem on the right, the sequence with the largest **joint probability** is “ok ok EOS”



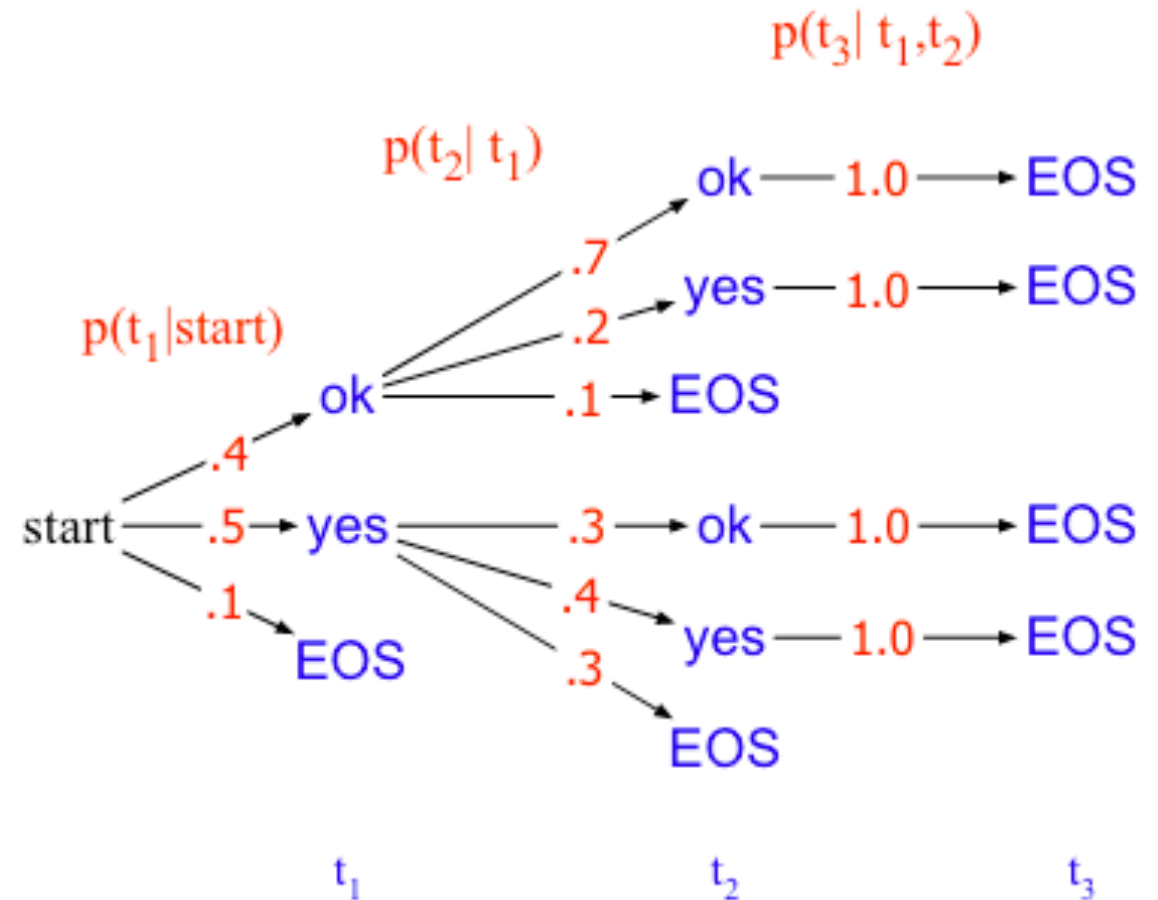
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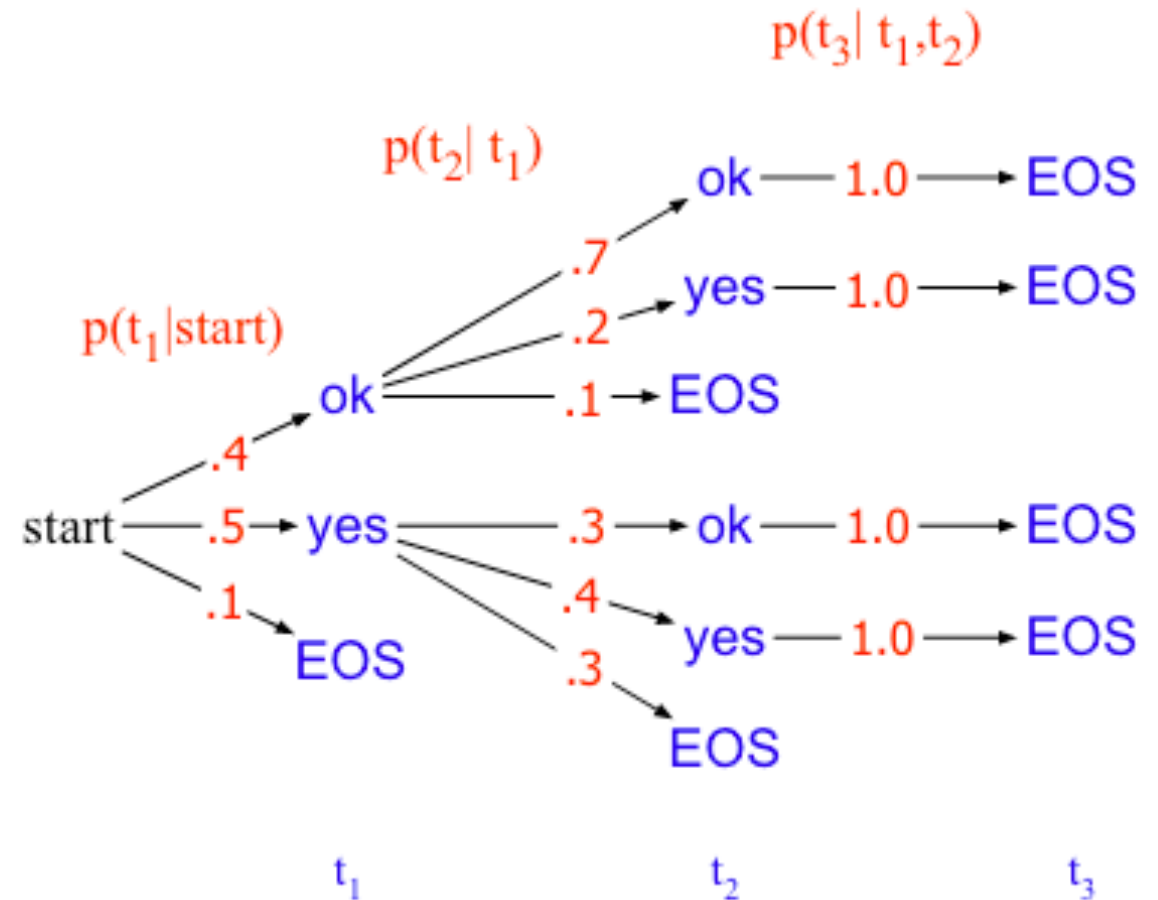
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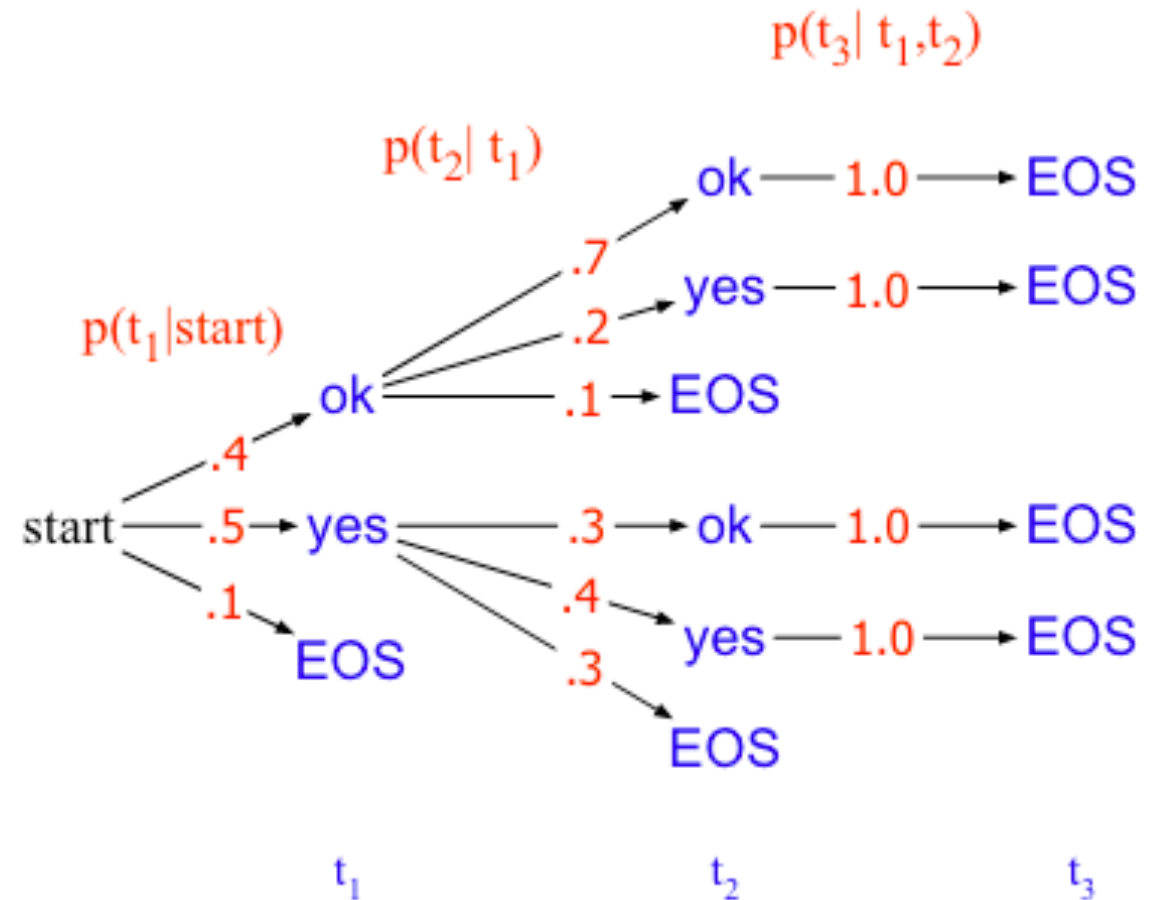
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 - Can you see why?



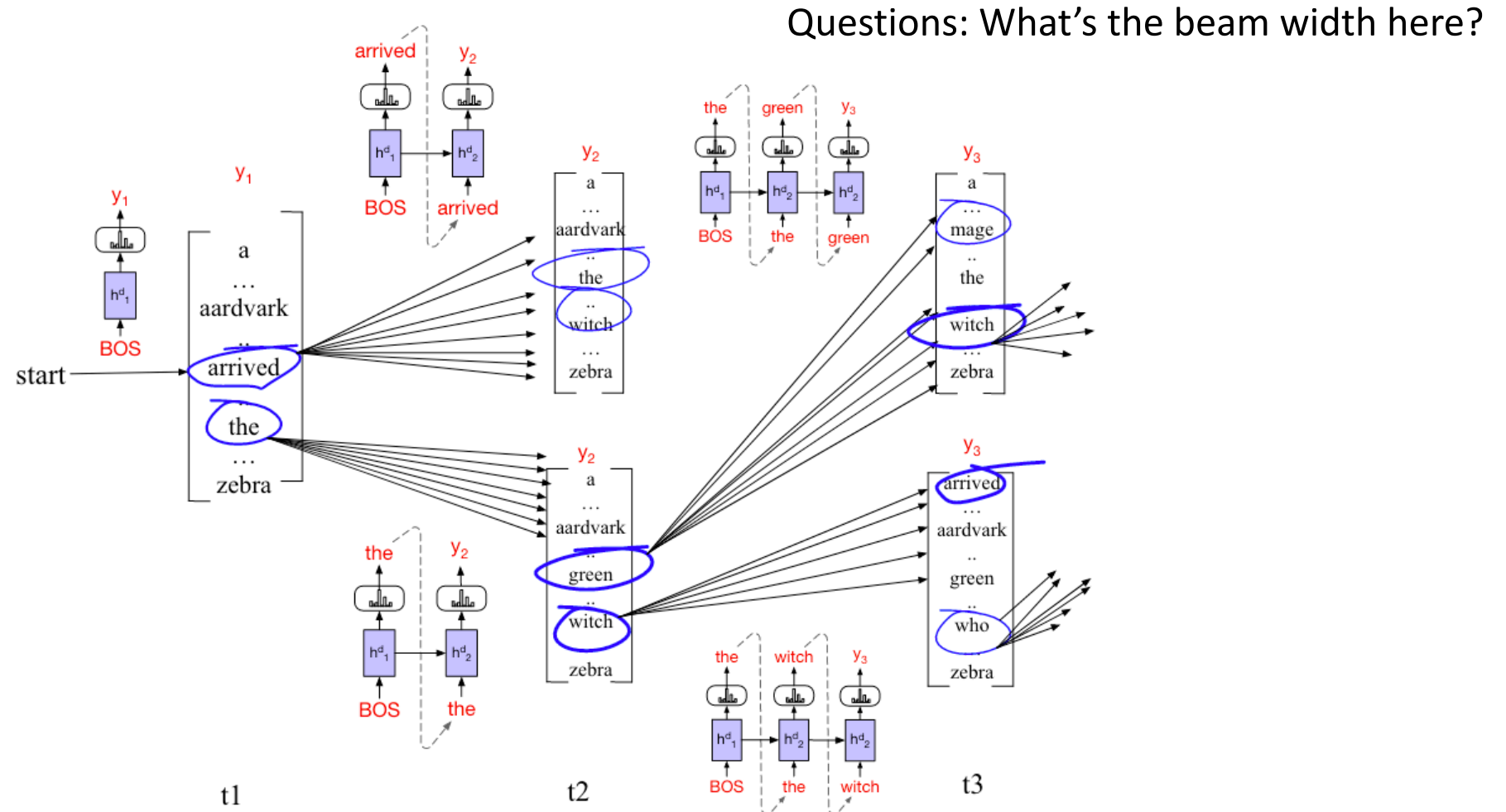
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- This is a beam search with $k=3$ beams
- Greedy-decoding is unable to decode that sequence
 - Can you see why?
 - What about beam search?



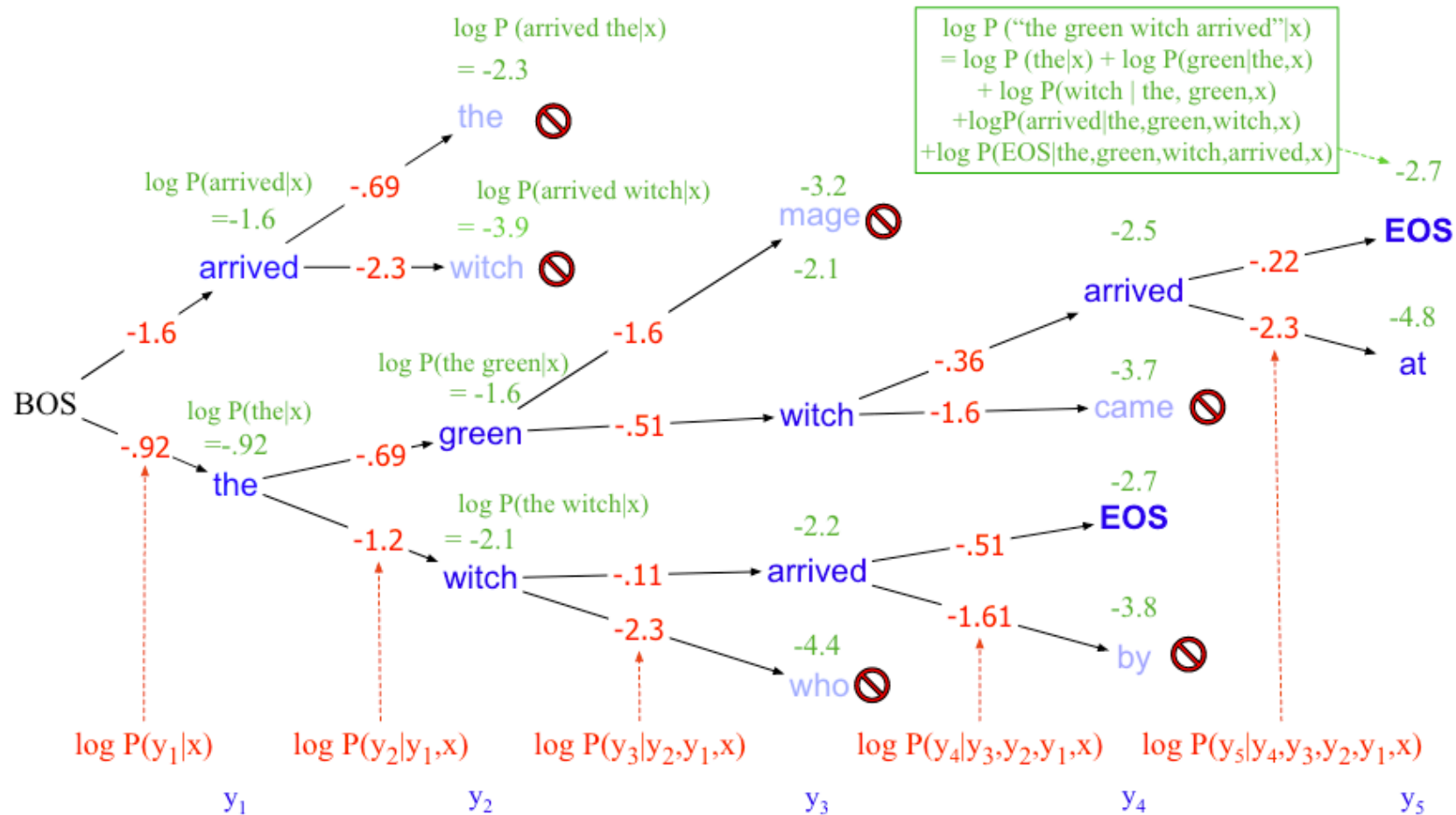
Decoding via beam-search

- At each step, consider k possible continuations (beam width)



Scoring with Beam search

- We maintain the log probability at each step and sum them along the paths we expand



Evaluating MT systems

- What makes a good translation?
 - **Adequacy**
 - **Fluency**
- The gold standard ~> **Human evaluation**
 - Problem: can be expensive and slow
- We want a good **automatic evaluation**
 - Potentially less accurate but hopefully still provide a good signal to compare two systems

Evaluating MT systems

- **Character F-score (chrF)**

- chrP: percentage of character 1-grams, 2-grams, ..., n-grams in the hypothesis that also occur in the target (averaged)
- chrR: percentage of character 1-grams, 2-grams, ..., n-grams in the target that also occur in the hypothesis (averaged)
- Compute an F-score by combining the two

$$\text{chrF}\beta = (1 + \beta^2) \frac{\text{chrP} \cdot \text{chrR}}{\beta^2 \cdot \text{chrP} + \text{chrR}}$$

$$\text{chrF2} = \frac{5 \cdot \text{chrP} \cdot \text{chrR}}{4 \cdot \text{chrP} + \text{chrR}}$$

Evaluating MT systems

- **BLEU score**
 - Purely precision based
 - N-gram word precision
 - Consider a corpus $\mathcal{C}$ with a single sentence, i.e., $\mathcal{C} = \{s\}$ (*Sentence BLEU*)
 - Uni-gram precision: percentage of uni-grams in the candidate translation (hypothesis) than also occur in the corpus

Evaluating MT systems

- ***Sentence BLEU score***
 - Compute n-gram precision for uni-, bi-, tri-, and 4-grams and take the geometric mean
- Downsides
 - Word based, making it hard to compare systems with different tokenization
 - Doesn't work well for languages with complex morphology
- Everyone in NLP knows that BLEU isn't great but we still use it...

Embedding based methods

Embedding based methods

chrF and BLEU both require a gold reference translation, what do we do when we don't have one?

Embedding based methods

chrF and BLEU both require a gold reference translation, what do we do when we don't have one?

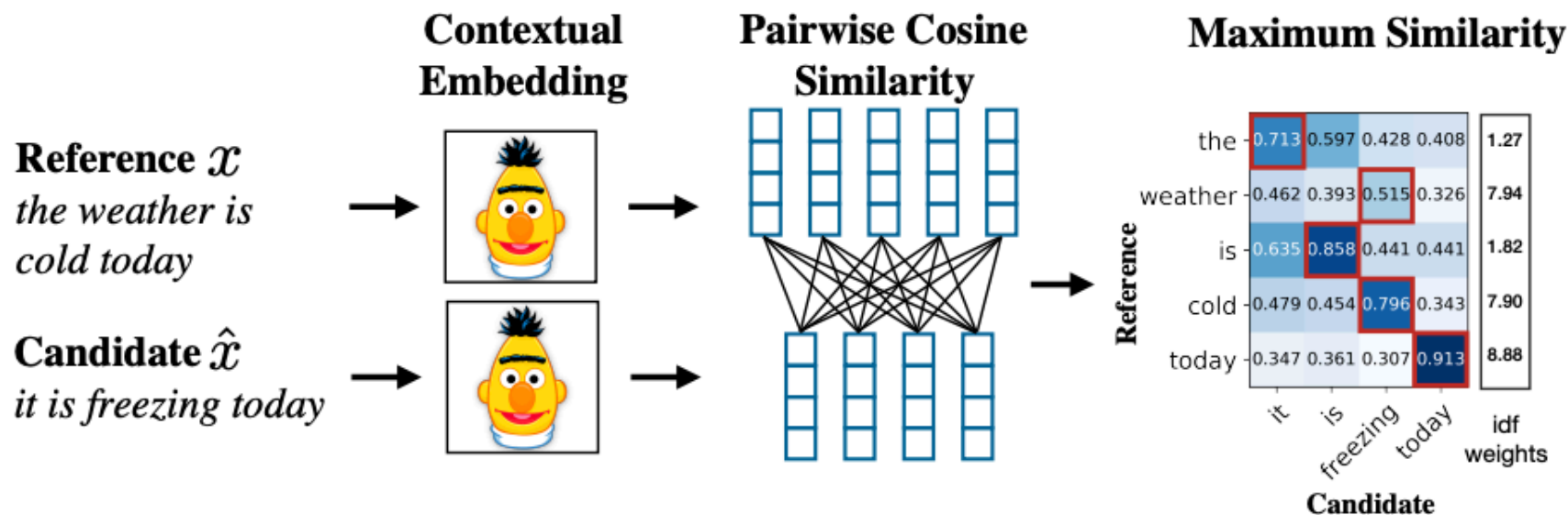
-> Embedding similarity!

Embedding based methods

chrF and BLEU both require a gold reference translation, what do we do when we don't have one?

-> Embedding similarity!

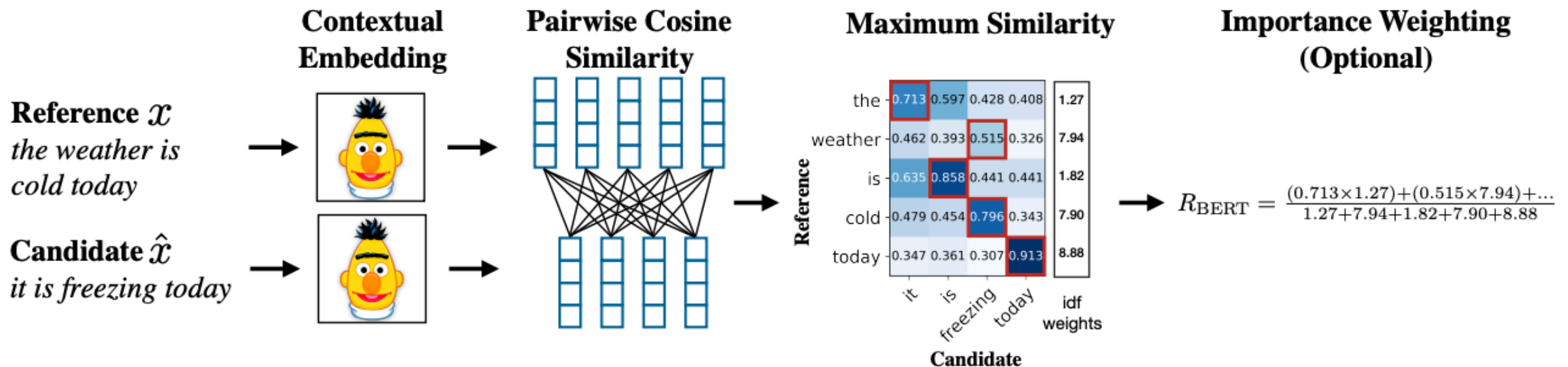
- Step 1 : Compare word embeddings from both the reference and translated sentences



Embedding based methods

- Step 2: Take max similarity pairs and get average across (i) reference words and (ii) candidate/translation words:

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\tilde{x}_j \in \tilde{x}} x_i \cdot \tilde{x}_j \quad P_{\text{BERT}} = \frac{1}{|\tilde{x}|} \sum_{\tilde{x}_j \in \tilde{x}} \max_{x_i \in x} x_i \cdot \tilde{x}_j$$



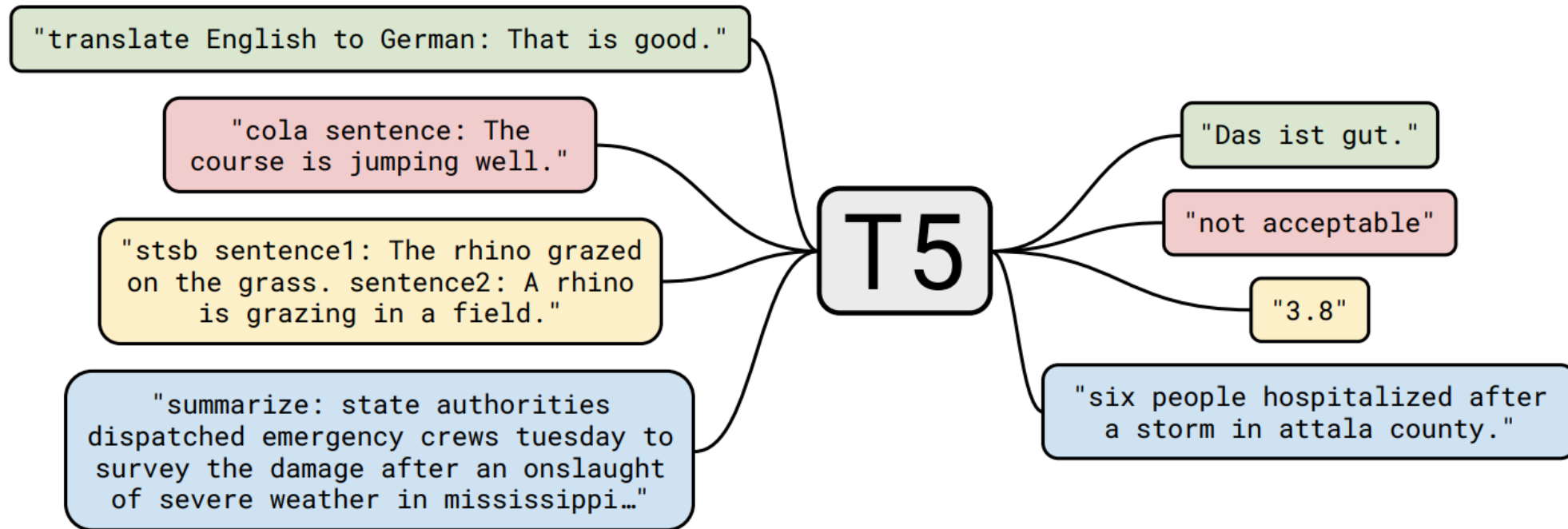
Bias and ethical issues

- Biases are often amplified in translation systems... because we always try to map to the highest probability output.
- What happens when we translate a language with a neutral pronoun (Hungarian) or that drops pronouns (Spanish) into a language with M/F pronouns (English or French)?

Hungarian (gender neutral) source	English MT output
ő egy ápoló	she is a nurse
ő egy tudós	he is a scientist
ő egy mérnök	he is an engineer
ő egy pék	he is a baker
ő egy tanár	she is a teacher
ő egy veskűvőszervező	she is a wedding organizer
ő egy vezérigazgató	he is a CEO

(Prates et al. 2019)

T5: seq-to-seq everywhere



There are a few famous figures in our field, this is one of them :)

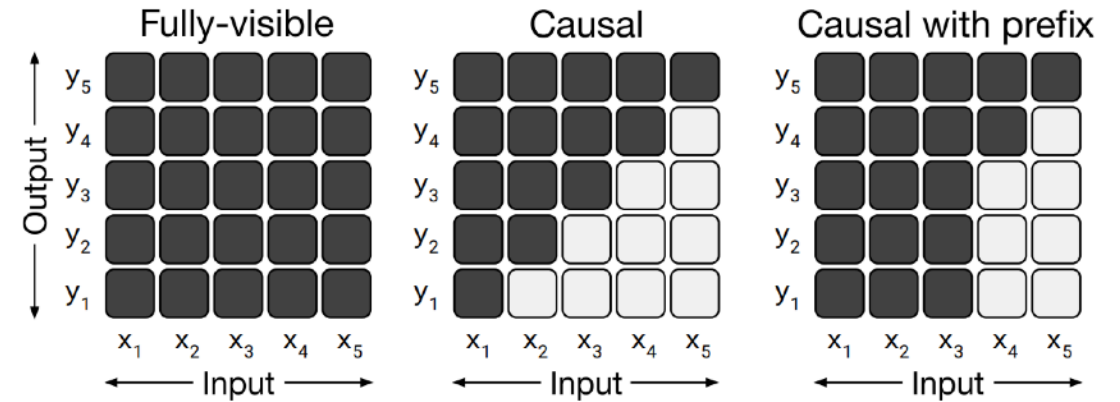
T5: seq-to-seq everywhere

- General **text-to-text** format
- Machine translation
 - Translate this text to French: Montréal is great. Montréal est génial
- Summarization
 - Summarize this news article: [Some article] [Summary of the article]
- What other tasks can you think of?
- Can you think of a NLP task that **can't** be formatted as text-to-text?

What's the best possible model for text-to-text learning?

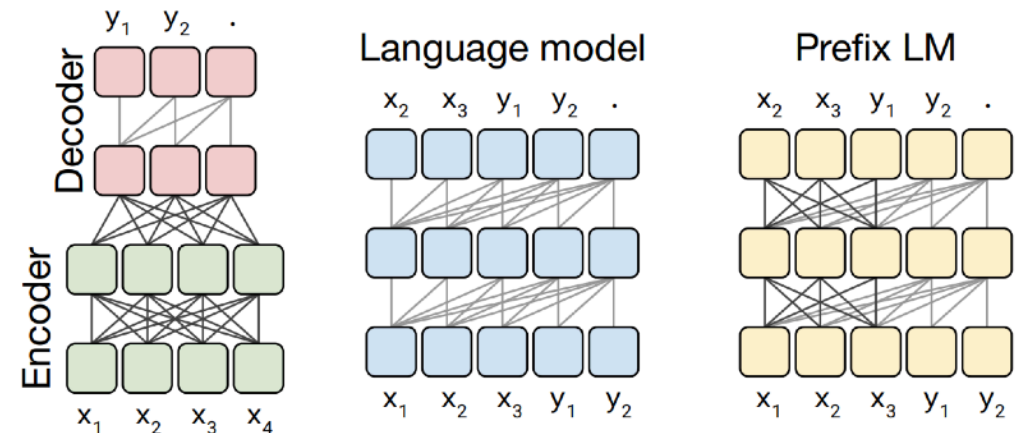
- **Model structure**

- Which attention should we use?
 - Bi-directional
 - Causal
 - A mix of the two



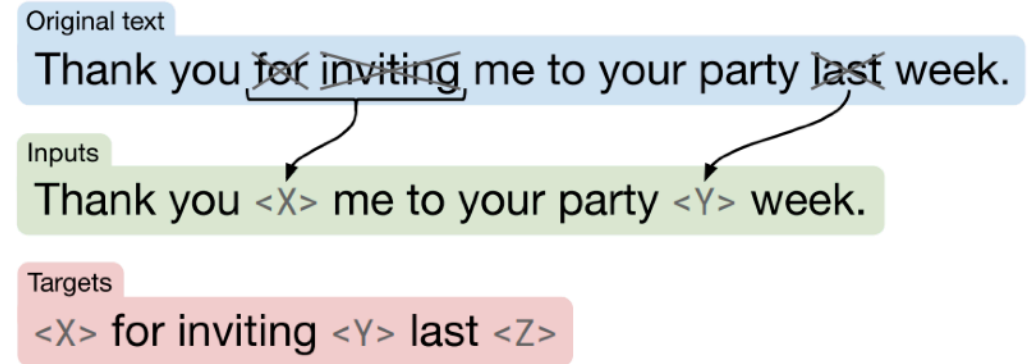
- **Model architecture**

- Encoder-decoder
- Decoder-only
- A mix of the two



What's the best possible model for text-to-text learning?

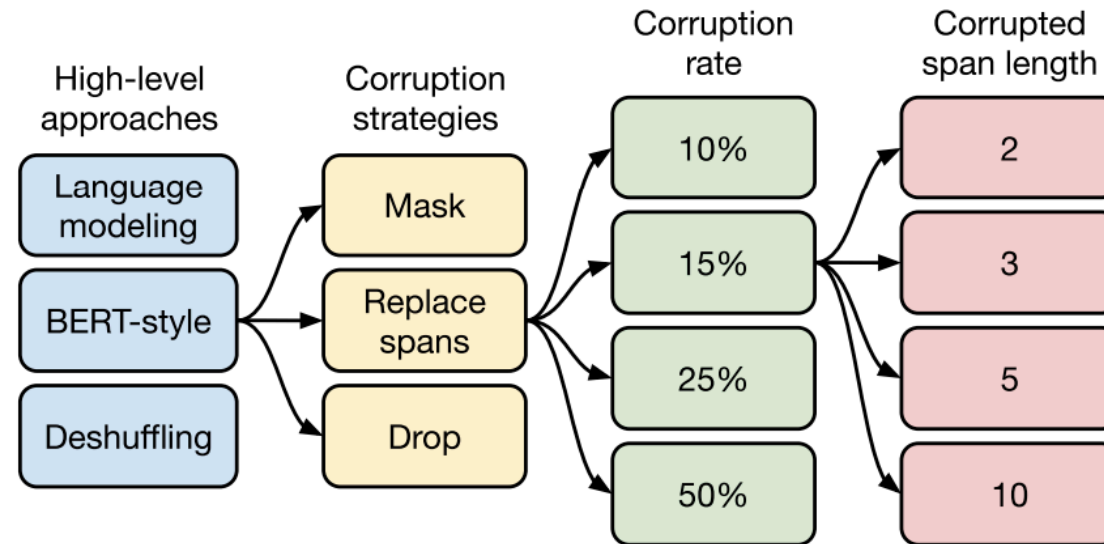
- **Training objective**
- What's the best way to encode useful information about and decode from a sequence?



Objective	Inputs	Targets
Prefix language modeling	Thank you for inviting	me to your party last week .
BERT-style Devlin et al. (2018)	Thank you <M> <M> me to your party apple week .	(original text)
Deshuffling	party me for your to . last fun you inviting week Thank	(original text)
MASS-style Song et al. (2019)	Thank you <M> <M> me to your party <M> week .	(original text)
I.i.d. noise, replace spans	Thank you <X> me to your party <Y> week .	<X> for inviting <Y> last <Z>
I.i.d. noise, drop tokens	Thank you me to your party week .	for inviting last
Random spans	Thank you <X> to <Y> week .	<X> for inviting me <Y> your party last <Z>

What's the best possible model for text-to-text learning?

- Simply try all possible combinations



What's the best possible model for text-to-text learning?

Objective	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
BERT-style (Devlin et al., 2018)	82.96	19.17	80.65	69.85	26.78	40.03	27.41
MASS-style (Song et al., 2019)	82.32	19.16	80.10	69.28	26.79	39.89	27.55
★ Replace corrupted spans	83.28	19.24	80.88	71.36	26.98	39.82	27.65
Drop corrupted tokens	84.44	19.31	80.52	68.67	27.07	39.76	27.82

Corruption rate	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
10%	82.82	19.00	80.38	69.55	26.87	39.28	27.44
★ 15%	83.28	19.24	80.88	71.36	26.98	39.82	27.65
25%	83.00	19.54	80.96	70.48	27.04	39.83	27.47
50%	81.27	19.32	79.80	70.33	27.01	39.90	27.49

Span length	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
★ Baseline (i.i.d.)	83.28	19.24	80.88	71.36	26.98	39.82	27.65
2	83.54	19.39	82.09	72.20	26.76	39.99	27.63
3	83.49	19.62	81.84	72.53	26.86	39.65	27.62
5	83.40	19.24	82.05	72.23	26.88	39.40	27.53
10	82.85	19.33	81.84	70.44	26.79	39.49	27.69

- Each of these matters to some extent

What's the best possible model for text-to-text learning?

Architecture	Objective	Params	Cost	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
★ Encoder-decoder	Denoising	$2P$	M	83.28	19.24	80.88	71.36	26.98	39.82	27.65
Enc-dec, shared	Denoising	P	M	82.81	18.78	80.63	70.73	26.72	39.03	27.46
Enc-dec, 6 layers	Denoising	P	$M/2$	80.88	18.97	77.59	68.42	26.38	38.40	26.95
Language model	Denoising	P	M	74.70	17.93	61.14	55.02	25.09	35.28	25.86
Prefix LM	Denoising	P	M	81.82	18.61	78.94	68.11	26.43	37.98	27.39
Encoder-decoder	LM	$2P$	M	79.56	18.59	76.02	64.29	26.27	39.17	26.86
Enc-dec, shared	LM	P	M	79.60	18.13	76.35	63.50	26.62	39.17	27.05
Enc-dec, 6 layers	LM	P	$M/2$	78.67	18.26	75.32	64.06	26.13	38.42	26.89
Language model	LM	P	M	73.78	17.54	53.81	56.51	25.23	34.31	25.38
Prefix LM	LM	P	M	79.68	17.84	76.87	64.86	26.28	37.51	26.76

- Some combinations are better than others
- Denoising (eg. MLM) seems to be almost always better than LMing with dec-enc architectures

(Raffel et al. 2020)

What's the best possible model for text-to-text learning?

- One important ingredient is missing
- Can you guess which one?

What's the best possible model for text-to-text learning?

- One important ingredient is missing
- Can you guess which one?

Data



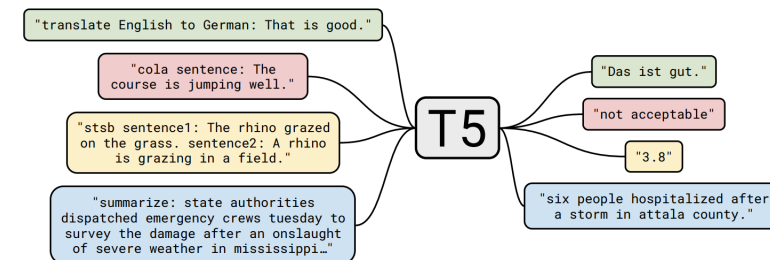
What's the best possible model for text-to-text learning?

- Data 
 - C4: Colossal Clean Crawled Corpus

Data set	Size	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
★ C4	745GB	83.28	19.24	80.88	71.36	26.98	39.82	27.65
C4, unfiltered	6.1TB	81.46	19.14	78.78	68.04	26.55	39.34	27.21
RealNews-like	35GB	83.83	19.23	80.39	72.38	26.75	39.90	27.48
WebText-like	17GB	84.03	19.31	81.42	71.40	26.80	39.74	27.59
Wikipedia	16GB	81.85	19.31	81.29	68.01	26.94	39.69	27.67
Wikipedia + TBC	20GB	83.65	19.28	82.08	73.24	26.77	39.63	27.57

Number of tokens	Repeats	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
★ Full data set	0	83.28	19.24	80.88	71.36	26.98	39.82	27.65
2^{29}	64	82.87	19.19	80.97	72.03	26.83	39.74	27.63
2^{27}	256	82.62	19.20	79.78	69.97	27.02	39.71	27.33
2^{25}	1,024	79.55	18.57	76.27	64.76	26.38	39.56	26.80
2^{23}	4,096	76.34	18.33	70.92	59.29	26.37	38.84	25.81

Overall performance

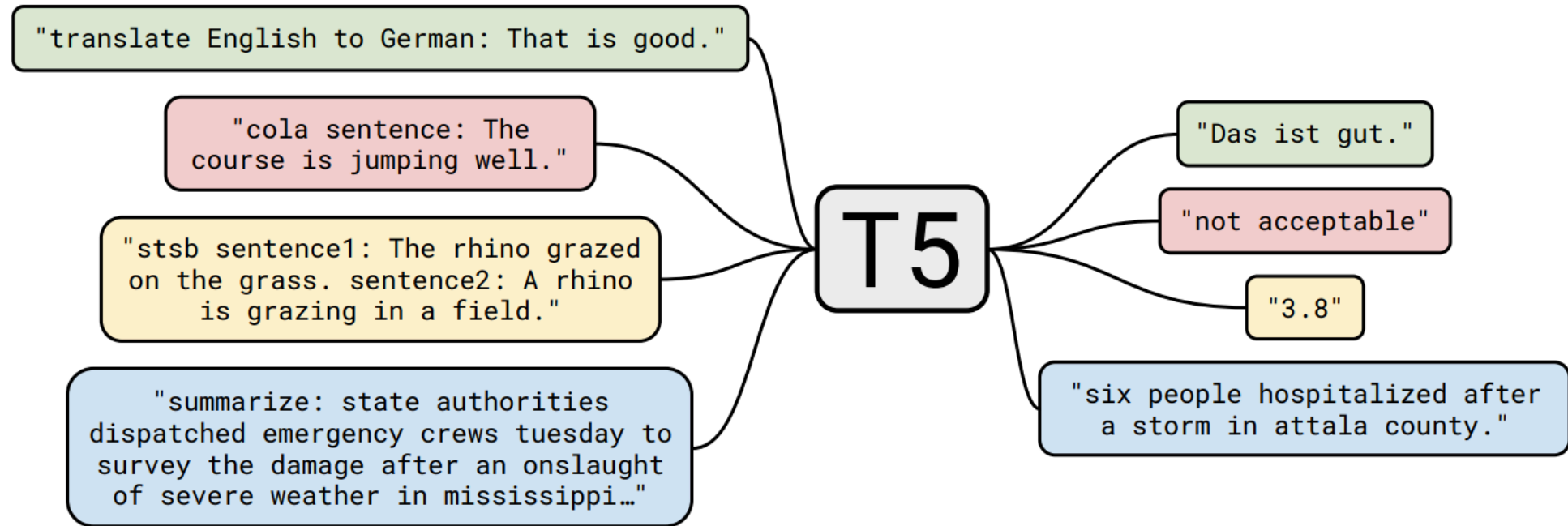


Model	GLUE Average	CoLA Matthew's	SST-2 Accuracy	MRPC F1	MRPC Accuracy	STS-B Pearson	STS-B Spearman
Previous best	89.4 ^a	69.2 ^b	97.1 ^a	93.6^b	91.5^b	92.7 ^b	92.3 ^b
T5-Small	77.4	41.0	91.8	89.7	86.6	85.6	85.0
T5-Base	82.7	51.1	95.2	90.7	87.5	89.4	88.6
T5-Large	86.4	61.2	96.3	92.4	89.9	89.9	89.2
T5-3B	88.5	67.1	97.4	92.5	90.0	90.6	89.8
T5-11B	90.3	71.6	97.5	92.8	90.4	93.1	92.8

Model	QQP F1	QQP Accuracy	MNLI-m Accuracy	MNLI-mm Accuracy	QNLI Accuracy	RTE Accuracy	WNLI Accuracy
Previous best	74.8 ^c	90.7^b	91.3 ^a	91.0 ^a	99.2^a	89.2 ^a	91.8 ^a
T5-Small	70.0	88.0	82.4	82.3	90.3	69.9	69.2
T5-Base	72.6	89.4	87.1	86.2	93.7	80.1	78.8
T5-Large	73.9	89.9	89.9	89.6	94.8	87.2	85.6
T5-3B	74.4	89.7	91.4	91.2	96.3	91.1	89.7
T5-11B	75.1	90.6	92.2	91.9	96.9	92.8	94.5

- Really good across the board. New SOTA on several tasks for *smaller* LMs.

T5: seq-to-seq everywhere



- Exhaustive search over the space of possible seq-to-seq models
- Very strong open source enc-dec model for NLP downstream tasks

T5: seq-to-seq everywhere

"translate English to German: That is good."

**But LLMs can do all of these to!
When might we prioritize one over the other?**

"summarize: state authorities
dispatched emergency crews tuesday to
survey the damage after an onslaught
of severe weather in mississippi..."

six people hospitalized after
a storm in attala county."

- Exhaustive search over the space of possible seq-to-seq models
- Very strong open source enc-dec model for NLP downstream tasks

Prompting: all tasks as text-to-text generation

Recall from Week 8

Prompt: a text string that a user issues to a language model to get the model to do something useful by conditional generation

Prompt engineering: the process of finding effective prompts for a task.

Prompting and in-context learning (ICL)

Recall from Week 8

- A prompt refers to the context provided to the language model
- Various components of a prompt, e.g., instruction, description, etc.

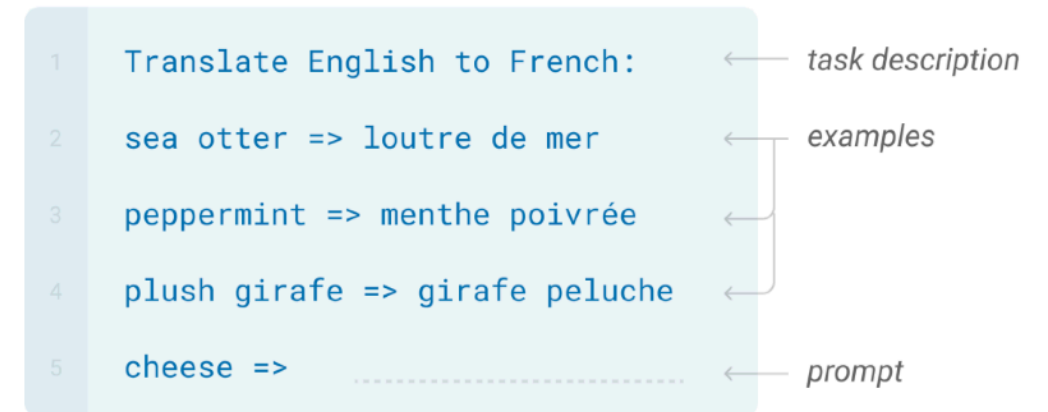
Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.



Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



[15 minute break]

Case study: An insurance buddy

Last week's case study

Your client is an insurance company. Evaluating claims fairly takes a lot of training and accumulated knowledge, but the company faces a lot of employee turnover (insurance isn't always the happiest of jobs!) which means that this knowledge keeps getting lost and training new employees is costly. As a solution to cut down on knowledge transfer time, you propose to create an AI assistant that will help claim agents with an initial assessment. It's role will be to identify (1) the most likely type of claim; (2) the sections in the client's policy that refer to this type of claim. It should then (3) generate an initial assessment of eligibility of the claim and possible resolutions. You have access to the following documents: (i) clients written claims and their full policies (mapped already), (ii) the set of all resolved claims and their resolutions from the past 5 years, (iii) a list of claim categories.

Case study: An insurance buddy

Last week's case study

- Sketch out the procedural structure of your AI Assistant, outlining each step and what NLP technologies will be necessary. If multiple are possible (eg. different LMs could be used) highlight which you would choose and why?
- Identify the limitations of your solutions and where they may introduce errors. What may be your level of responsibility and accountability for these errors?
- **Consider possible challenges that your solution may present to data sovereignty, client privacy, and compute costs to the company.**

The Clarifying Lawful Overseas Use of Data (CLOUD) Act, passed in 2018, allows U.S. law enforcement to compel U.S.-based technology companies to disclose electronic data, such as emails or files, regardless of whether the data is stored in the United States or on foreign servers. It focuses on control rather than data location. [Wikipedia +3](#)

Key Aspects of the CLOUD Act

- **Scope:** Applies to any service provider subject to U.S. jurisdiction, covering major cloud providers like Google, Amazon, Microsoft, and Apple.
- **Cross-Border Access:** Eliminates the need to rely solely on slow, traditional Mutual Legal Assistance Treaties (MLAT) for accessing data stored abroad.
- **Data Sovereignty Conflicts:** It can force companies to turn over data even if doing so violates local privacy laws (e.g., GDPR in the EU or Canadian law).

[News](#) > [World](#) > [Europe](#)

European governments ditch US tech services over security fears

European governments are moving away from US tech giants, opting for domestic or open-source alternatives

France is ditching American tech. When will Canada?



VASS BEDNAR >

SPECIAL TO THE GLOBE AND MAIL

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What do we do when the client is a bank, a government agency, or a company whose value is their data?

Case study: An insurance buddy continued

Phase 1 of your AI assistant was well received and you are now asked to extend it to their Quebec market. You have to contend with documents sometimes being in English and other times in French. Insurance policy holders are always provided their policy document in French in Quebec, but the company keeps copies of many of these policies in English as well, though not all of them, since many of the claims agents are located outside Quebec and do not speak French. As with last week, the role of your agent is still to: identify (1) the most likely type of claim and (2) the sections in the client's policy that refer to this type of claim; and to then (3) generate an initial assessment of eligibility of the claim and possible resolutions. Additionally, **the agent should (4) generate a translation of its assessment in French to be provided to clients with French claims.** As for documents: **now (i) client claims can also be in French and their mapped full policies as well, and about 30% of these policies have existing readily available translations (there are no translations for French claims though).** (ii) The set of all resolved claims and their resolutions from the past 5 years and (iii) the list of claim categories are all in English.

Case study: An insurance buddy continued

- Sketch out the procedural structure of your AI Assistant, outlining each step and what NLP technologies will be necessary. If multiple are possible (eg. different LMs could be used) highlight which you would choose and why?
- Explain what data preprocessing steps will be necessary for your solution
- Identify the limitations of your solution and where they may introduce errors. What may be your level of responsibility and accountability for these errors?
- Explain how you plan to evaluate your system during the development phase and afterwards.